

Tweaking the constant in the Linear Hadwiger Theorem

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Abstract

Norin and Steiner recently proved the Linear Hadwiger Conjecture. That is, there is an absolute constant C such that every graph G satisfies $\chi(G) \leq C \text{ had}(G)$, where $\chi(G)$ is the chromatic number and $\text{had}(G)$ is the Hadwiger number of G . Their proof gives a non-optimised constant C of order 10^{100} . This paper uses extensive AI-based optimisation to show that every graph G satisfies $\chi(G) \leq 19,885,160 \text{ had}(G)$.

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1 Introduction

A graph H is a *minor* of a graph G if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges. The *Hadwiger number* $\text{had}(G)$ is the largest integer t such that the complete graph K_t is a minor of G . A *colouring* of G assigns one colour to each vertex so that adjacent vertices receive distinct colours. The *chromatic number* $\chi(G)$ is the minimum number of colours in a colouring of G . In 1943 Hadwiger [14] proposed the following conjecture. It remains one of the deepest open problems in graph theory.

Conjecture 1.1 (Hadwiger’s Conjecture [14]). *Every graph G satisfies $\chi(G) \leq \text{had}(G)$.*

The conjecture is trivial for $\text{had}(G) \leq 2$. It was proved for $\text{had}(G) = 3$ by Hadwiger [14] and Dirac [10]. For $\text{had}(G) = 4$ it is equivalent to the Four Colour Theorem [44, 1, 2, 38, 15, 16]. For $\text{had}(G) = 5$ it was reduced to the Four Colour Theorem by Robertson, Seymour and Thomas [39]. It is open for $\text{had}(G) \geq 6$. See the surveys of Seymour [40] and Norin [33] for background.

The exact conjecture appears to be out of reach. So much effort has gone into upper bounds on $\chi(G)$ as a function of $t = \text{had}(G)$. Table 1 summarises the history.

Year	Upper bound on $\chi(G)$ when $\text{had}(G) = t$	Reference
1964	2^{t-1}	Wagner [45]
1968	$O(t \log t)$	Mader [28]
1982/84	$O(t\sqrt{\log t})$	Kostochka [21, 22], Thomason [41]
2001	$(2\alpha + o(1)) t\sqrt{\log t}$	Thomason [42]
2019	$O(t(\log t)^\beta)$ for every $\beta > \frac{1}{4}$	Norin, Postle and Song [35]
2021	$O(t \log \log t)$	Delcourt and Postle [8]
2026	$O(t\sqrt{\log \log t})$	Liu and Luo [26], Lin [25]
2026	$O(t \log \log \log t)$	Liu and Luo [27]
2026	$O(t)$, with constant of order 10^{100}	Norin and Steiner [36]
2026	$19,885,160 t$	this paper

Table 1: Upper bounds on the chromatic number of graphs with Hadwiger number t . Years refer to the first appearance of each result. Here $\alpha = 0.3190863\dots$ is Thomason’s constant.

The first bounds came from degeneracy. A graph is *d-degenerate* if every non-null subgraph has a vertex of degree at most d . A d -degenerate graph is $(d+1)$ -colourable, by a greedy argument. Note that $\text{had}(H) \leq \text{had}(G)$ for every subgraph H of G . So an upper bound on the average degree of a graph in terms of its Hadwiger number gives an upper bound on its degeneracy, and hence on its chromatic number. Let G be a graph with $\text{had}(G) = t \geq 2$. Mader [28] showed that G has average degree $O(t \log t)$. Kostochka [21, 22] and Thomason [41] independently improved this to $O(t\sqrt{\log t})$. Thomason [42] later determined the extremal function asymptotically. He defined an explicit constant $\alpha = 0.3190863\dots$ and showed that every graph G with no K_t minor has at most $(\alpha + o(1)) t\sqrt{\log t} v(G)$ edges. So G has average degree at most $(2\alpha + o(1)) t\sqrt{\log t}$. This gives the bounds from 1968 to 2001 in Table 1.

The bound $O(t\sqrt{\log t})$ on the average degree is tight, as shown by random graphs. Bollobás, Catlin and Erdős [5] proved that the random graph $G(n, \frac{1}{2})$ almost surely has Hadwiger number $(1+o(1)) n/\sqrt{\log_2 n}$. So for each t there is a graph with no K_t minor and average degree $\Omega(t\sqrt{\log t})$. Thomason [42] showed that the constant α is best possible, with random graphs as extremal examples, and observed that the extremal graphs are essentially disjoint unions of dense pseudo-random graphs. Myers [31] made this precise. Hence no argument based only on degeneracy can do better than $\Theta(t\sqrt{\log t})$. This is the *degeneracy barrier*.

However, these random graphs are far from being counterexamples to Hadwiger’s Conjecture. Indeed, $G(n, \frac{1}{2})$ almost surely has chromatic number $(1 + o(1))n/(2 \log_2 n)$, by a theorem of Bollobás [4], which is much smaller than its Hadwiger number. Evidence of a different kind comes from fractional colouring. Reed and Seymour [37] proved that $\chi_f(G) \leq 2 \text{had}(G)$ for every graph G , where χ_f is the fractional chromatic number. So the fractional relaxation of Hadwiger’s Conjecture holds up to a factor of 2. This suggested the *Linear Hadwiger Conjecture*, which asserts that $\chi(G) \leq C \text{had}(G)$ for some absolute constant C .

The degeneracy barrier stood for more than thirty years. Norin, Postle and Song [35] were the first to break the degeneracy barrier. They proved that graphs with no K_t minor are $O(t(\log t)^\beta)$ -colourable for every $\beta > \frac{1}{4}$. Roughly speaking, their approach shows that a graph with no K_t minor and large chromatic number contains a small highly connected subgraph, and that many such subgraphs can be linked together to form a K_t minor. A key tool is their density increment theorem, which finds in a graph either a denser minor or a small subgraph of nearly the same density. A density bound of Norin and Postle [34] for unbalanced bipartite graphs with no K_t minor was used later by Delcourt and Postle [8] to find small dense subgraphs.

Delcourt and Postle [8] then proved the bound $O(t \log \log t)$. They also proved a reduction of a different nature. If every graph with no K_t minor and $O(t \log^4 t)$ vertices is $O(t)$ -colourable, then the Linear Hadwiger Conjecture holds. So only graphs of order $t(\log t)^{O(1)}$ need to be considered. For such graphs, the Duchet–Meyniel bound $\alpha(G) \geq n/(2t)$ [11] and greedy colouring give $\chi(G) = O(t \log(n/t)) = O(t \log \log t)$, where n is the number of vertices. This is how Delcourt and Postle obtained their bound. Lin [24] later reduced the order of the graphs in this reduction to $O(t \log^2 t)$.

After the reduction of Delcourt and Postle, the problem was to colour small graphs with no K_t minor more efficiently. Graphs with $n = O(t)$ vertices are trivially $O(t)$ -colourable. So the difficulty lies with graphs whose order is at most $t(\log t)^{O(1)}$. Liu and Luo [26] and, independently, Lin [25] improved the colouring of such graphs, and obtained the bound $O(t\sqrt{\log \log t})$. Liu and Luo [27] then improved this to $O(t \log \log \log t)$.

Shortly afterwards, Norin and Steiner [36] proved the Linear Hadwiger Conjecture. They colour graphs with at most $t(\log t)^{1/2-\varepsilon}$ vertices directly, using the theorem of Reed and Seymour, the Rödl nibble and a linear programming argument. They then bootstrap up to graphs with $t(\log t)^5$ vertices, and finish with the reduction of Delcourt and Postle.

Theorem 1.2 (Norin and Steiner [36]). *There is an absolute constant C such that every graph G satisfies $\chi(G) \leq C \text{had}(G)$.*

Norin and Steiner [36, Section 4] estimate that their proof gives a non-optimised constant C of order 10^{100} . They write:

“There is a lot of room for improvement, and we see no clear barriers to reducing the value of C below 10^{10} . Proving Theorem 2 with, say, $C = 100$ would likely require substantial new ideas.”

Here their Theorem 2 is Theorem 1.2. Our main result verifies their first remark.

Theorem 1.3. *Every graph G satisfies $\chi(G) \leq 19,885,160 \text{had}(G)$.*

The author does not claim that Theorem 1.3 is a significant improvement on the result of Norin and Steiner (which is a real breakthrough). The present paper simply reports the outcomes of an experiment in using AI to optimise the constants.

Statement about AI use. The author started by asking Claude Opus 5.5 to optimise the proof of Norin and Steiner [36]. Claude identified several areas for optimisation, worked out all the details, and produced a first draft of the paper. GPT-6 checked the proof and suggested further optimisations. The author also identified optimisations, and instructed Claude how to improve the exposition. This process went through several iterations. The author understands the proof at a high level, but has not checked all the details by hand. Since the proof is technical and largely numerical, the author considers computer verification to be at least as reliable as human verification.

Formal verification. Theorem 1.3 has been formally verified in the proof assistant Lean 4 [23] with the Mathlib library [30]. The formalisation builds on the Lean formalisation by Norin and Steiner [36] of their proof and of its external inputs. On top of this, Claude Opus 5.5 formalised the arguments of the present paper, with the differences listed below, together with the following external results that we use beyond those of Norin and Steiner: Thomas and Wollan’s linkage theorems with the constant 5 [43, Theorem 1.6 and Corollary 1.2], the degree condition of Kawarabayashi, Kostochka and Yu [18, Theorem 1.1] on which they rely, and Nguyen’s theorem [32, Theorem 1.2]. The final Lean statement is exactly Theorem 1.3. The formalisation differs from the paper in a few inessential places. The size thresholds in Lemma 10.3, and in Lemma 10.4 and Proposition 10.5, are $\log \log a \geq 1000$ and $\log \log a \geq 1001$ rather than 879.8 and 881, because the base case is derived from a formalised quantitative theorem of Norin and Steiner rather than from Lemmas 10.1 and 10.2. Lemma 10.6 uses the bound $v(J) \leq 2\alpha(J)\text{had}(J)$, which follows from the theorem of Reed and Seymour [37] and is formalised by Norin and Steiner, in place of Theorem 4.13. All constants are the same. The Lean files are available as ancillary files with the arXiv version of this paper.

The remainder of the paper is organised as follows. Section 2 introduces the main constants used. Section 3 explains how the argument of Norin and Steiner is modified. Section 4 gives definitions and the results we use. Section 5 derives a weaving lemma from Wollan’s work. Section 6 proves the refined form of Lin’s theorem. Sections 7, 8 and 9 contain the three main ingredients. Section 10 makes all size thresholds explicit, combines the three ingredients in a final separation step (Section 10.4), and proves Theorem 1.3. Appendix A discusses how the constant might be reduced further.

2 Constants

A constant introduced in (the statement or proof of) Theorem, Lemma, Proposition or Corollary X.Y is denoted by a letter with subscript X.Y, for example $C_{5.4}$ in Lemma 5.4. The one exception is the sequence $c^{(L)}$ of Corollary 8.2, which is indexed by the number L of bootstrapping steps. Auxiliary parameters used only locally, such as the stage factor $M = 10^5$ and the parameters inside proofs, are not covered by this convention. Constants from other papers are named after the result of this paper in which they are restated, and each restatement cites its source; the only exception is Section 3, which uses the notation of Norin and Steiner. We use C for constants in edge-density bounds, Λ for coefficients in lower bounds on the chromatic number that force a structure, c for coefficients in upper bounds on the chromatic number, τ for a bound on the order of a subgraph, and ω for a proportion of density.

The constant in Theorem 1.3 is

$$c_{1.3} := (1 + 10^{-6})(7030.6 + 2C_{9.1} + \frac{472}{3}c_{10.5}) < 19,885,160.$$

Here $c_{10.5} = 126,115.1$ is the constant for colouring graphs of order $t(\log t)^{O(1)}$ (Proposition 10.5 and Corollary 10.7), and $C_{9.1} = 18,000$ is the constant in the density threshold

$d = C_{9.1}t$ of the chromatic-inseparable case (Proposition 9.1). The term $\frac{472}{3}c_{10.5}$ accounts for 99.8% of the constant; Appendix A explains where each term comes from. The factor $1 + 10^{-6}$ converts a bound in terms of $\text{had}(G) + 1$ into a bound in terms of $\text{had}(G)$.

Table 2 lists the main constants.

Constant	Value	Role	Where
$\omega_{6.2}$	0.0044	proportion of density kept by the refined Lin theorem	Thm 6.2
$\tau_{6.2}$	$4.56 \cdot 10^{18}$	pieces have at most $\tau_{6.2} t \log t$ vertices	Thm 6.2
M	10^5	each stage adds $\lceil t/(M\sqrt{\log t}) \rceil$ branch sets	Lemma 5.4
$C_{5.4}$	70	density forcing the weaving property	Lemma 5.4
$C_{9.1}$	18,000	density threshold $d = C_{9.1}t$, with $\omega_{6.2}C_{9.1} \geq C_{5.4} + 9$	Prop. 9.1
$c_{10.4}$	988.9	additive cost of one bootstrapping step	Lemma 10.4
$c_{10.5}$	126,115.1	colouring graphs with at most N_a vertices	Prop. 10.5
$\Lambda_{10.9}$	$\approx 1.934 \cdot 10^7$	$(1 + \eta)(897.5 + \frac{230}{3}(78 + 2c_{10.5}))$, final reduction	Lemma 10.9
$c_{1.3}$	$< 19,885,160$	$(1 + 10^{-6})(7030.6 + 2C_{9.1} + \frac{472}{3}c_{10.5})$	Thm 1.3

Table 2: The main constants.

3 From the Norin–Steiner argument to the present bound

We first recall the structure of the proof of Norin and Steiner [36]. Let $\text{LH}(\alpha)$ denote the following statement. There is a constant C_α such that, for all sufficiently large t , every graph G with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^\alpha$ satisfies $\chi(G) \leq C_\alpha t$. The proof has three parts.

- (I) *Small graphs.* Norin and Steiner [36, Theorem 4] prove that $\text{LH}(\alpha)$ holds with constant $4 + \varepsilon$ for every $\alpha < \frac{1}{2}$. This uses the fractional colouring theorem of Reed and Seymour [37], a quantitative form of the Pippenger–Frankl–Rödl theorem on almost perfect matchings in hypergraphs, proved by Norin and Steiner [36, Theorem 9 and Appendix A], and a linear programming argument.
- (II) *Bootstrapping.* Norin and Steiner [36, Theorem 5] prove that $\text{LH}(\alpha)$ implies $\text{LH}(\frac{4}{3}\alpha)$, with new constant $2C'$, where $C' := 2C_\alpha + C_{\text{sep}}(1 + C_\alpha)$. Their proof gives C' for integer t , and the factor 2 extends it to real t . Here C_{sep} is the constant in a reduction to chromatic-inseparable subgraphs due to Norin and Steiner [36, Theorem 15], adapted from Delcourt and Postle [8].
- (III) *Large graphs.* By a theorem of Delcourt and Postle [8, Theorem 1.6], $\text{LH}(5)$ implies Theorem 1.2. This costs a further multiplicative constant.

Part (II) is applied ten times to pass from exponent $\frac{1}{3}$ to exponent 5. Here, in the notation of Norin and Steiner, $C_{\text{sep}} = 39 \cdot 35 \cdot C_{B.1}^2 \cdot C_{B.8}$, by Norin and Steiner [36, Appendix B.3 and the proof of Lemma B.12]. Here $C_{B.1}$ is the density constant of the small-dense-subgraph theorem of Delcourt and Postle (Theorem 4.12 below, where it is $C_{4.12} = 3,072,000$). Norin and Steiner only need $C_{B.1} \geq 480$. Also $C_{B.8}$ is the linkage constant of Bollobás and Thomason [6] in the form of [36, Theorem B.8], which Norin and Steiner only require to satisfy $C_{B.8} \geq 1$. By the theorem of Thomas and Wollan [43, Corollary 1.3], we may take $C_{B.8} = 10$. Even with $C_{B.1} = 480$ we get $C_{\text{sep}} \geq 39 \cdot 35 \cdot 480^2 \cdot 10 \approx 3.1 \cdot 10^9$, and ten iterations of $C \mapsto 2(2C + C_{\text{sep}}(1 + C))$ starting from 5 give a constant of at least about $5.8 \cdot 10^{98}$. With $C_{B.1} = 3,072,000$ it would be about 10^{175} . Part (III) then multiplies this by another large constant. So the estimated constant 10^{100} is the result of three multiplicative compoundings.

We now summarise how the present paper improves upon the proof of Norin and Steiner [36].

1. An explicit chromatic-separable reduction (Section 7). We reprove the reduction of Norin and Steiner [36, Theorem 15] with explicit constants. We use a theorem of Nguyen [32] on highly connected subgraphs with large chromatic number (Theorem 4.2). It replaces the theorem of Girão and Narayanan [12]. We also use the linkage theorem of Thomas and Wollan [43]. We obtain the linking hub by trimming the chromatic number (an idea from the proof of [36, Lemma B.12]), passing to a vertex-critical subgraph, and extracting a dense highly connected subgraph using a sharpened form of Mader’s theorem (Lemmas 6.5 and 7.1). The density form of the Thomas–Wollan theorem then makes the hub linked, at chromatic cost about 12.5 per linkage pair, compared with about 30.6 via connectivity alone. This avoids a small-dense-subgraph theorem. We also distribute the branch sets unevenly among the three subgraphs used in the recursion, giving more of them to the subgraph that costs only one separation. This lowers the coefficient of Φ but raises the constant term, so we use whichever distribution is better for the given Φ . The result (Theorem 7.3) replaces $C_{\text{sep}}(1+\Phi)$ by $42 + \min\{517.5 + 84(1+\Phi), 897.5 + \frac{230}{3}(1+\Phi)\}$.

2. An additive bootstrap (Section 8). Each bootstrapping step of Norin and Steiner splits the graph into two parts. The part G_1 is covered by large connected induced bipartite subgraphs. The part G_2 has no such subgraph. The reduction of Part (II) is applied to G_2 . The chromatic-inseparable subgraphs of G_2 are handled by a Gyárfás path argument together with the result of the previous level. This makes the constants multiply. We instead always handle the Gyárfás graphs with the base result of Part (I). Then each step costs only a factor 2 (from the bipartite contraction) plus an additive constant (Theorem 8.1). Seven steps lead from exponent just below $\frac{1}{2}$ to exponent above 2.

3. The chromatic-inseparable case (Section 9). Delcourt and Postle [8, Lemma 2.5] show that if a graph with no K_t minor is $Ct(1+g)$ -chromatic-inseparable, then its chromatic number is less than $2Ct(1+g)$. Here C is a large constant, and g measures the chromatic number of graphs of order $t(\log t)^{O(1)}$. So the large constant appears both in the cost of separating and in the lower bound on the chromatic number. We rework their proof (Proposition 9.1). Let $\Gamma := tg$. The proof shows that a graph with no K_t minor and chromatic number at least $(32+2C_{9,1})t+2\Gamma+2$ is $(78t+2\Gamma)$ -chromatic-separable. In particular, every s -chromatic-inseparable graph with no K_t minor satisfies $s < (2C_{9,1} + 33)t + 2\Gamma$. In both forms, the large constant $C_{9,1}$ multiplies only t , and not the small-graph term Γ . Moreover, the large constant $C_{9,1}$ enters only the chromatic threshold, that is, the lower bound on the chromatic number that forces separability, and not the cost of separating. In the final application (Section 10.4) the chromatic threshold is paid once, as an additive term carried through the induction of the chromatic-separable reduction, while each separation costs only about $(78 + 2c_{10.5})$ times the Hadwiger number. The final constant is therefore $2C_{9,1}$ plus a multiple of the small-graph constant, and not a product.

4. Rooted minors via massed pairs (Section 5). Inside the chromatic-inseparable case, small highly connected pieces must contain rooted complete minors together with disjoint linkages. Delcourt and Postle obtain these from Wollan’s extremal theorem for rooted minors [46, Theorem 1.1], applied to $K_s + \ell K_2$ with $\ell \approx 10t$. This requires density of order $10^5 t$. We use instead two structural results from Wollan’s proof, concerning a minimal counterexample (Theorem 5.1), together with the strategy of his proof of [46, Theorem 5.1]. A minimal counterexample contains a dense graph J of order $O(t)$. Since

J is dense and small, a random set of at most $0.008t$ of its vertices contains a K_s model, and the roots and the matching pairs can be linked around this set using the density form of the Thomas–Wollan linkage theorem. So J has the required rooted models, which contradicts minimality (Lemma 5.3). This gives density $70t$ (Lemma 5.4). We also arrange that the result is applied to the piece itself, with no vertices deleted, so that only its density matters.

5. A refined form of Lin’s theorem (Section 6). The chromatic-inseparable case uses a theorem that finds small highly connected subgraphs in dense graphs with no K_t minor. Delcourt and Postle [8, Theorem 2.3] find such subgraphs with $O(t \log^3 t)$ vertices at density constant about $3.07 \cdot 10^6$. Lin [24] reduced the order to $O(t \log t)$, but with density constant about $1.31 \cdot 10^9$. We rerun Lin’s argument with a much larger bound Q on the weight of the contracted subgraphs, and with balanced parameters. We also use a simple averaging lemma and the proof of Mader’s theorem to obtain pieces that are dense, and not only highly connected (Theorem 6.2). This matters because Wollan’s theorem needs density, while the connectivity needed in the chromatic-inseparable case is only $9t$. A graph of density $d \geq 1.8 \cdot 10^4 t$ with no K_t minor contains a $(9t + 1)$ -connected piece of density more than $\omega_{6.2}d - 9t$, where $\omega_{6.2} = 0.0044$. The order bound grows to about $4.6 \cdot 10^{18} t \log t$. This affects only the size thresholds (lower bounds on t), not the final constant. Because the pieces have order $O(t \log t)$, the small graphs that must be coloured have $O(t \log^2 t)$ vertices. So seven bootstrapping steps suffice instead of fifteen.

6. A more economical construction (Sections 5 and 9). In the chromatic-inseparable case, each highly connected piece must contain a rooted complete minor together with a linkage. Delcourt and Postle allow $10t$ linkage pairs. At stage T of the construction the number of pairs is $(4T - 3)x \leq (4 + o(1))t$, where x is the number of branch sets added at each stage. We use $\lfloor 4.004t \rfloor$ pairs. Delcourt and Postle take $x = \lceil t/\sqrt{\log t} \rceil$, so that the rooted clique K_{2x} needs density about $60t$ by Kostochka’s theorem. We take $x = \lceil t/(M\sqrt{\log t}) \rceil$ with $M = 10^5$, which makes this term negligible, at the cost of M times as many stages. This affects only the size thresholds, and it is what makes the clique model in Lemma 5.3 small. Finally, the pieces are knitted together inside the hub using the density form of the Thomas–Wollan linkage theorem (Lemma 4.6), which needs connectivity only about $7t$ instead of about $40t$.

7. Explicit size thresholds (Section 10). Our asymptotic statements hold for t larger than explicit size thresholds, which are lower bounds on t . In the bootstrap the size threshold is $\log \log t \geq 881$. Below it, graphs of order $t(\log t)^{O(1)}$ are coloured using the Duchet–Meyniel bound on the independence number together with greedy colouring (Lemma 10.6), which gives a bound much smaller than the bootstrap constant there. In the final step the size threshold is $\log \log t \geq 25$, and below it the degeneracy bound of Theorem 4.11 gives at most $1.7 \cdot 10^7 t$, which is less than our constant. The estimate of Norin and Steiner [36, Section 4] instead uses the degeneracy bound for moderate t , at a cost of order $\sqrt{\log t_0}$ for a size threshold t_0 . In an explicit version of their argument, this adds a further large factor to the three compoundings described above.

4 Preliminaries

4.1 Definitions

All graphs are finite and simple. For an integer $n \geq 0$, let $[n] := \{1, \dots, n\}$, so $[0] = \emptyset$. All logarithms are natural. For $k \geq 1$, we write $\log^k t := (\log t)^k$. For a graph G , let $v(G)$

denote the number of vertices of G . Let $e(G)$ denote the number of edges. The *order* of G is $v(G)$. The *density* of a non-null graph G is $e(G)/v(G)$, which is half its average degree. For $v \in V(G)$, let $N_G(v)$ be the set of neighbours of v in G , and let $\deg_G(v) := |N_G(v)|$. Let $\delta(G)$ and $\Delta(G)$ denote the minimum and maximum degree. For an integer $k \geq 0$, a graph is *k-connected* if it has more than k vertices and remains connected after deleting any set of fewer than k vertices. For a non-null graph G , the *connectivity* $\kappa(G)$ is the largest k such that G is k -connected. So G is k -connected if and only if $\kappa(G) \geq k$, and $\kappa(G) \leq \delta(G)$ whenever $v(G) \geq 2$. A subgraph H of G is *spanning* if $V(H) = V(G)$. For $X \subseteq V(G)$, let $G[X]$ denote the subgraph induced by X . Let $G - X := G[V(G) \setminus X]$. For disjoint $A, B \subseteq V(G)$, let $e_G(A, B)$ be the number of edges of G with one end in A and the other in B , and let $G(A, B)$ be the bipartite subgraph of G formed by these edges. For graphs G_1, G_2 , the graph $G_1 \cup G_2$ has vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2)$. The *disjoint union* $G_1 + G_2$ is the union of vertex-disjoint copies of G_1 and G_2 . For an integer $\ell \geq 0$, let ℓG denote the disjoint union of ℓ copies of G . The *null graph* has no vertices.

Contraction. Let $A \subseteq V(G)$ be non-empty with $G[A]$ connected. The graph G/A is obtained from $G - A$ by adding a new vertex x_A adjacent to every vertex of $V(G) \setminus A$ that has a neighbour in A . We say that G/A is obtained by *contracting* A . For pairwise disjoint such sets $A_1, \dots, A_m$, contracting each A_i means applying these operations one after another. The result does not depend on the order. Contracting an edge uv means contracting $\{u, v\}$. Contracting a connected subgraph H means contracting $V(H)$. All contractions produce simple graphs. Finally, for distinct vertices u, v , which need not be adjacent, *identifying* u and v to a new vertex c means deleting u and v and adding c adjacent to every vertex of $(N_G(u) \cup N_G(v)) \setminus \{u, v\}$.

Colourings. A *colouring* of G is a map $c : V(G) \rightarrow \mathbb{N}$ with $c(u) \neq c(v)$ for every edge uv . The chromatic number $\chi(G)$ is the minimum number of colours used by a colouring. An *independent set* is a set of pairwise non-adjacent vertices. The *independence number* $\alpha(G)$ is the maximum size of an independent set. The *fractional chromatic number* $\chi_f(G)$ is the minimum of $\sum_I x_I$ over all non-negative weights x_I on the independent sets I of G such that $\sum_{I \ni v} x_I \geq 1$ for every vertex v . Summing these constraints over all vertices shows that $v(G) \leq \chi_f(G) \alpha(G)$. A graph is *d-degenerate* if every non-null subgraph has a vertex of degree at most d . A d -degenerate graph satisfies $\chi(G) \leq d + 1$.

Minors and models. A graph H is a *minor* of G if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges. The *Hadwiger number* $\text{had}(G)$ is the largest integer t such that K_t is a minor of G , with $\text{had}(G) := 0$ if G is null. A *model* of a graph H in G is a family $\{B_v : v \in V(H)\}$ of pairwise disjoint subsets of $V(G)$ such that each $G[B_v]$ is connected and there is an edge of G between B_u and B_v for every edge uv of H . The sets B_v are called *branch sets*. A graph G has an H minor if and only if it contains a model of H . Let $X = \{x_v : v \in V(H)\}$ be a set of $|V(H)|$ distinct vertices of G . The model is *rooted at* X (according to the bijection $x_v \mapsto v$) if $x_v \in B_v$ for every v . For a K_a model and a set R of a vertices, *rooted at* R means rooted at R according to some bijection between R and $V(K_a)$. Equivalently, each branch set contains exactly one vertex of R . A K_s *model* is a model of K_s , that is, a family of s pairwise adjacent branch sets. If $\mathcal{A}$ is a model, then $V(\mathcal{A})$ denotes the union of its branch sets. Similarly, for a family $\mathcal{P}$ of paths, $V(\mathcal{P}) := \bigcup_{P \in \mathcal{P}} V(P)$. A subgraph or family is *disjoint from* a set X if its vertex set is disjoint from X , and two subgraphs are *disjoint* if they are vertex-disjoint. Paths are *pairwise disjoint outside* X if any two of them have no common vertex outside X .

Separations. A *separation* of G is a pair (A, B) of subsets of $V(G)$ with $A \cup B = V(G)$ and no edge between $A \setminus B$ and $B \setminus A$. Its *order* is $|A \cap B|$. It is *trivial* if $A \subseteq B$ or $B \subseteq A$.

Paths and linkages. A *path* $P = v_0 v_1 \dots v_\ell$ in G is a sequence of distinct vertices with

$v_{i-1}v_i \in E(G)$ for each $i \in [\ell]$. We identify P with the subgraph it forms. Its *ends* are v_0 and v_ℓ . We call v_0 its *first* vertex and v_ℓ its *last* vertex. The other vertices are *internal*. A path P is *induced* in G if $G[V(P)] = P$. Every subpath of an induced path is induced, and an induced path can be properly coloured with two colours. Given any path in G with ends u and v , a shortest u - v path in $G[V(P)]$ is an induced path of G with the same ends, and its vertex set is contained in $V(P)$. For $A, B \subseteq V(G)$, an A - B *path* is a path whose first vertex is its only vertex in A and whose last vertex is its only vertex in B . Let $s_1, \dots, s_\ell, t_1, \dots, t_\ell$ be vertices such that $s_i \neq s_j$ and $t_i \neq t_j$ for $i \neq j$, and such that $s_i = t_j$ only if $i = j$. An $\{(s_i, t_i)\}_{i \in [\ell]}$ -*linkage* is a family of pairwise vertex-disjoint paths $P_1, \dots, P_\ell$ such that P_i has ends s_i and t_i . Following Thomas and Wollan [43], a graph is ℓ -*linked* if it has at least 2ℓ vertices and such a linkage exists for every choice of 2ℓ distinct vertices $s_1, \dots, s_\ell, t_1, \dots, t_\ell$. Every linkage used with Theorem 4.5 below has 2ℓ distinct ends.

Woven graphs. The following notion is due to Delcourt and Postle [8, Definition 5.12]. A graph G is (a, b) -*woven* if the following holds for all sets $R = \{r_1, \dots, r_a\}$, $S = \{s_1, \dots, s_b\}$ and $T = \{t_1, \dots, t_b\}$ of vertices with $s_i = t_j$ only if $i = j$. There are a K_a model $\mathcal{M}$ rooted at R and an $\{(s_i, t_i)\}$ -linkage $\mathcal{P}$ with $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. Here the vertices $r_1, \dots, r_a$ are distinct, and so are $s_1, \dots, s_b$ and $t_1, \dots, t_b$. Wovenness is monotone in the following sense. This is used implicitly in the proof of a lemma of Delcourt and Postle [8, Lemma 5.15], and it is the reason for the hypothesis $v(H) \geq a + 2b$ in the version of that lemma given by Norin and Steiner [36, Lemma B.11].

Lemma 4.1. *Let $a \geq a' \geq 0$ and $b \geq b' \geq 0$ be integers. If G is (a, b) -woven and $v(G) \geq a + 2b$, then G is (a', b') -woven.*

Proof. Let $R' = \{r_1, \dots, r_{a'}\}$, $S' = \{s_1, \dots, s_{b'}\}$ and $T' = \{t_1, \dots, t_{b'}\}$ be as in the definition of (a', b') -woven. Since $|R' \cup S' \cup T'| \leq a' + 2b'$ and $v(G) \geq a + 2b$, there are $(a - a') + 2(b - b')$ distinct vertices of G outside $R' \cup S' \cup T'$. Call them $r_{a'+1}, \dots, r_a, s_{b'+1}, \dots, s_b$ and $t_{b'+1}, \dots, t_b$. These new vertices are *padding*. Let $R := \{r_1, \dots, r_a\}$, $S := \{s_1, \dots, s_b\}$ and $T := \{t_1, \dots, t_b\}$. Then $s_i = t_j$ only if $i = j$, because this holds for $i, j \leq b'$ and every padding vertex is distinct from all other vertices of $R \cup S \cup T$. Since G is (a, b) -woven, there are a K_a model $\{M_1, \dots, M_a\}$ with $r_i \in M_i$ and an $\{(s_i, t_i)\}_{i \in [b]}$ -linkage $\{P_1, \dots, P_b\}$ with $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. Let $\mathcal{M}' := \{M_1, \dots, M_{a'}\}$ and $\mathcal{P}' := \{P_1, \dots, P_{b'}\}$. Then $\mathcal{M}'$ is a $K_{a'}$ model rooted at R' , and $\mathcal{P}'$ is an $\{(s_i, t_i)\}_{i \in [b']}$ -linkage. Clearly $R' \cap (S' \cup T') \subseteq V(\mathcal{M}') \cap V(\mathcal{P}')$. Conversely, let $v \in V(\mathcal{M}') \cap V(\mathcal{P}')$, say $v \in M_i$ with $i \leq a'$ and $v \in V(P_j)$ with $j \leq b'$. Then $v \in R \cap (S \cup T)$. Since $v \in R$, we have $v = r_{i'} \in M_{i'}$ for some i' , and $i' = i$ because the branch sets are disjoint. So $v = r_i \in R'$. Since $v \in S \cup T$, the vertex v is an end of the path of $\mathcal{P}$ for its own pair, and the paths are disjoint. So that path is P_j , and $v \in \{s_j, t_j\} \subseteq S' \cup T'$. Hence $V(\mathcal{M}') \cap V(\mathcal{P}') = R' \cap (S' \cup T')$. $\square$

Knit graphs. The following notion is due to Bollobás and Thomason [6]. We use the form of Delcourt and Postle [8, Definition 5.7], without the requirement $\ell \leq k \leq v(G)$. A graph G is (k, ℓ) -*knit* if the following holds for every set S of k vertices and every partition of S into $p \geq \ell$ non-empty parts $S_1, \dots, S_p$. There are pairwise disjoint sets $D_1, \dots, D_p$ with $S_i \subseteq D_i$ and each $G[D_i]$ connected.

Chromatic separability. The following notion is due to Delcourt and Postle [8, Definition 2.4]. Let $s \geq 0$. A graph G is s -*chromatic-separable* if it has vertex-disjoint subgraphs H_1, H_2 , possibly null, with $\chi(H_i) \geq \chi(G) - s$ for $i = 1, 2$. Otherwise G is s -*chromatic-inseparable*. Allowing null subgraphs is implicit in the work of Norin and Steiner [36, Section 3.1], and differs from [8] only for very small graphs. Delcourt and Postle take s to be an integer, and we allow real s . Taking $H_1 = G$ and H_2 null shows that every s -chromatic-inseparable graph satisfies $\chi(G) > s$. If G is s -chromatic-inseparable and $s' \leq s$,

then G is s' -chromatic-inseparable. If H_1, H_2 are as above, then $G[V(H_1)]$ and $G[V(H_2)]$ are vertex-disjoint induced subgraphs of G with $\chi(G[V(H_i)]) \geq \chi(H_i) \geq \chi(G) - s$. So G is s -chromatic-separable if and only if it has two vertex-disjoint induced subgraphs, possibly null, each with chromatic number at least $\chi(G) - s$.

Tangency and cores. The following notions are due to Delcourt and Postle [8, Definition 6.3]. A subgraph J of G is *tangent* to a model $\mathcal{A} = \{A_1, \dots, A_s\}$ if $|V(J) \cap A_i| = 1$ for every i . A set $S \subseteq V(G)$ is a *core* of $\mathcal{A}$ if, for all distinct i, j , some edge of G has one end in $A_i \cap S$ and the other in $A_j \cap S$. Such edges are called *core edges*.

Bipartite pieces. Following Norin and Steiner [36, Section 3], where it is denoted $b(G)$, let $\text{bip}(G)$ denote the maximum number of vertices in a connected induced bipartite subgraph of G .

The extremal function. For a positive integer s , let $D(s)$ denote the maximum chromatic number of a graph with no K_s minor. For real $y \geq 1$, let $D(y) := D(\lfloor y \rfloor)$.

The hypothesis $(\mathcal{H}_{\Phi, a_0})$. Let $\Phi \geq 0$ and $a_0 \geq 1$ be real. A graph G satisfies $(\mathcal{H}_{\Phi, a_0})$ if the following holds for every induced subgraph H of G . If H is s -chromatic-inseparable and $\text{had}(H) < b$ for some real $b \geq a_0$, then $s \leq \Phi b$. This property is inherited by induced subgraphs. It is adapted from the hypothesis on the function f in a theorem of Norin and Steiner [36, Theorem 15]. (The corresponding theorem of Delcourt and Postle [8, Theorem 7.1] uses a hypothesis of a different kind, namely an upper bound on $\chi(H)/a$ over small subgraphs H with no K_a minor.) The differences from the hypothesis of Norin and Steiner are that the lower bound a_0 on b is a free parameter, and that b is not bounded above.

4.2 Connectivity, linkages and minors

Theorem 4.2 (Nguyen [32, Theorem 1.2]). *Let $k \geq 1$ be an integer. Every graph F with $\chi(F) \geq (3 + \frac{1}{16})k$ contains a $(k+1)$ -connected subgraph with more than $\chi(F) - k$ vertices and chromatic number at least $\chi(F) - 2k + 2$.*

Nguyen's definition of k -connectivity agrees with ours. In the notation of Nguyen [32], Theorem 4.2 implies (and Nguyen's abstract states) that $g(k, m) \leq \max\{m + 2k - 2, \lceil (3 + \frac{1}{16})k \rceil\}$ for all integers $k \geq 1$ and $m \geq 2$, where $g(k, m)$ is the least n such that every graph with chromatic number at least n contains a $(k+1)$ -connected subgraph with chromatic number at least m .

We use Theorem 4.2 in the following form.

Corollary 4.3. *Let $k \geq 2$ be an integer and let F be a graph with $\chi(F) \geq \lceil 3.0625(k-1) \rceil$. Then F contains a k -connected induced subgraph F' with $\chi(F') \geq \chi(F) - 2k$.*

Proof. Apply Theorem 4.2 with $k-1 \geq 1$ in place of k . Since $\chi(F)$ is an integer, $\chi(F) \geq \lceil 3.0625(k-1) \rceil$ is equivalent to $\chi(F) \geq (3 + \frac{1}{16})(k-1)$. This gives a k -connected subgraph F_0 of F with $\chi(F_0) \geq \chi(F) - 2(k-1) + 2 = \chi(F) - 2k + 4 \geq \chi(F) - 2k$. The graph $F' := F[V(F_0)]$ contains F_0 as a spanning subgraph, so it is also k -connected and $\chi(F') \geq \chi(F_0)$. $\square$

We use Menger's theorem in the following standard form.

Theorem 4.4 (Menger). *Let $k \geq 1$ be an integer, let G be a k -connected graph, and let $A, B \subseteq V(G)$ be disjoint sets with $|A|, |B| \geq k$. Then G contains k pairwise vertex-disjoint A - B paths.*

Theorem 4.5 (Thomas and Wollan [43, Corollaries 1.2 and 1.3]). *Let $\ell \geq 1$ be an integer. Every 2ℓ -connected graph G with at least $5\ell v(G)$ edges is ℓ -linked. In particular, every 10ℓ -connected graph is ℓ -linked.*

Lemma 4.6 (cf. Delcourt and Postle [8, Theorem 5.8]). *Let ℓ and k be integers with $1 \leq \ell \leq k$. Every graph G with $\kappa(G) \geq 2k - \ell$ and $e(G) \geq 5k v(G)$ is (k, ℓ) -knit.*

Proof. Delcourt and Postle [8, Theorem 5.8] state a result of this kind for Ck -connected graphs, with an unspecified constant C , as a corollary of their theorem [8, Theorem 5.6], for whose linkage constant they remark that the Thomas–Wollan theorem gives the value 10. Since our graphs are very dense, we use the density form of Theorem 4.5, which needs much less connectivity. Let $|S| = k$, and let $S_1, \dots, S_p$ be a partition of S into non-empty parts, where $p \geq \ell$, with $S_i = \{v_{i,1}, \dots, v_{i,n_i}\}$. If $p = k$, then every part is a single vertex, and $D_i := S_i$ works. So let $p < k$. Since $\delta(G) \geq \kappa(G) \geq 2k - p$, we can choose pairwise distinct vertices $u_{i,j} \notin S$, for $1 \leq i \leq p$ and $2 \leq j \leq n_i$, such that $u_{i,j}$ is adjacent to $v_{i,j}$. Indeed, when $u_{i,j}$ is chosen, fewer than $k + (k - p)$ vertices are forbidden. Let $L := \{v_{i,n_i} : i \in [p]\}$. The pairs $(v_{i,j-1}, u_{i,j})$, for $i \in [p]$ and $2 \leq j \leq n_i$, number $k - p$, have pairwise distinct ends, and avoid L . The graph $G - L$ is $(\kappa(G) - p)$ -connected, and $\kappa(G) - p \geq 2k - \ell - p \geq 2(k - p)$. At most $p v(G)$ edges of G have an end in L , so $e(G - L) \geq (5k - p) v(G) \geq 5(k - p) v(G - L)$. By Theorem 4.5, $G - L$ is $(k - p)$ -linked, so it has a linkage for these pairs. Every vertex of $S \setminus L$ is an end of one of the pairs, so it lies on no other path, and the paths avoid L . So every vertex of S on a path is an end of that path. For each i , let D_i be the union of S_i and the vertex sets of the paths for the pairs with first index i . Then $G[D_i]$ is connected, since it contains these paths and the edges $u_{i,j}v_{i,j}$. Also $S_i \subseteq D_i$, and the sets D_i are pairwise disjoint. $\square$

Kawarabayashi and Yu [19] claimed that every 9ℓ -connected graph is ℓ -knitted, that is, every set of ℓ vertices can be knitted for every partition into non-empty parts, together with a density form. In a recent version [20] they correct a gap in that argument and prove that every 8ℓ -connected graph is ℓ -knitted. In our application the connectivity of the pieces is determined by Lemma 5.4 rather than by the knitting, so these results would not change our constants.

Lemma 4.7 (Delcourt and Postle [8, Lemma 5.3]). *Let G be a graph, and let A_1, A_2, B be disjoint non-empty subsets of $V(G)$. Suppose that, for each $i \in \{1, 2\}$, there is a set $\mathcal{P}_i$ of $2|A_i|$ A_i – B paths in $G - A_{3-i}$ that are pairwise disjoint in $G - A_i$, such that every vertex of A_i lies in exactly two paths of $\mathcal{P}_i$. Then G contains $|A_1| + |A_2|$ pairwise vertex-disjoint $(A_1 \cup A_2)$ – B paths.*

Lemma 4.8 (Delcourt and Postle [8, Lemma 5.15]). *Let $a \geq 1$ and $b \geq 0$ be integers. Let $\mathcal{P}$ be an $\{(s_i, t_i)\}_{i \in [b]}$ -linkage in a graph G , where $\{s_i\}$ and $\{t_i\}$ are disjoint. Let H be an (a, b) -woven subgraph of G with $v(H) \geq a + 2b$, and let $R \subseteq V(H)$ with $|R| = a$. Then G has an $\{(s_i, t_i)\}$ -linkage $\mathcal{P}'$ and a K_a model $\mathcal{M}$ in H rooted at R such that $V(\mathcal{P}') \cap V(\mathcal{M}) \subseteq R \cap V(\mathcal{P})$ and $V(\mathcal{P}') \subseteq V(H) \cup V(\mathcal{P})$.*

The statement of Delcourt and Postle [8] says that $\mathcal{M}$ is a model in G . The proof constructs it inside H . The proof applies the wovenness of H to at most b pairs. This uses Lemma 4.1, which is why the hypothesis $v(H) \geq a + 2b$ is needed. Norin and Steiner [36, Lemma B.11] include the same hypothesis.

Lemma 4.9 (Kawarabayashi [17, statement (3.1)], see also [8, Lemma 5.14]). *Let $a \geq 1$ be an integer and let G be a graph. If $\kappa(G) \geq a$ and G has a K_{2a} minor, then G is $(a, 0)$ -woven.*

Lemma 4.10 (Delcourt and Postle [8, Lemma 6.2]). *Let G be a connected graph and $\emptyset \neq S \subseteq V(G)$. Then there are an induced connected subgraph H of G and a set S' with $S \subseteq S' \subseteq V(H)$, $|S'| \leq 3|S|$ and $\chi(H - S') \leq 2$.*

4.3 Extremal results

Theorem 4.11 (Kostochka [21, 22], see also [8, Theorem 4.3]). *Let $s \geq 2$ be an integer. Every non-null graph G with $e(G) \geq 30s\sqrt{\log s} v(G)$ has a K_s minor. Consequently every graph with no K_s minor is $(60s\sqrt{\log s})$ -degenerate, and $D(s) \leq 60s\sqrt{\log s} + 1$.*

Delcourt and Postle derive Theorem 4.11 from Kostochka's theorem that every graph with n vertices and at least kn edges, where $k \geq 2$, has a K_h minor for every integer $h \leq 0.064k/\sqrt{\log k}$, with natural logarithms [21, Theorem; p. 16 of the translation] (see also [22, p. 309]). Indeed, with $k = 30s\sqrt{\log s}$, one checks that $0.064k/\sqrt{\log k} \geq s$ for all $s \geq 4$. For $s \in \{2, 3\}$ the conclusion is trivial, since a non-null graph G with $e(G) \geq 30s\sqrt{\log s} v(G) \geq 2v(G)$ contains a cycle, and hence a K_3 minor.

The next result is the small-dense-subgraph theorem of Delcourt and Postle.

Theorem 4.12 (Delcourt and Postle [8, Theorem 2.3]). *Let $C_{4.12} := 480 \cdot 6400 = 3,072,000$. For all integers $t \geq 3$ and $k \geq t$, every non-null graph G with no K_t minor and $e(G) \geq C_{4.12}k v(G)$ has a k -connected subgraph with at most $C_{4.12}^2 t \log^3 t$ vertices.*

In the work of Delcourt and Postle [8] the constant is $480 C_{6.3}$, where $C_{6.3}$ is the constant in an unbalanced bipartite density bound for graphs with no K_t minor, stated by Delcourt and Postle [8, Theorem 4.1]. That bound is due to Norin and Postle [34, Theorem 3.2], who prove it with constant 6400 (Theorem 6.3 below). Lin [24, Theorem 1.4] proved a version of Theorem 4.12 with pieces of order $O(t \log t)$. We do not use Theorem 4.12 itself. Instead, Section 6 proves a refined form of Lin's theorem, which finds pieces that are both dense and highly connected.

Theorem 4.13 (Duchet and Meyniel [11], see also [8, Theorem 3.3]). *Let $a \geq 2$ be an integer. Every graph J with no K_a minor satisfies $\alpha(J) \geq v(J)/(2(a-1))$.*

The next lemma restates results of Delcourt and Postle [8, Lemma 3.1 and Corollary 3.2] in terms of the independence number.

Lemma 4.14 (Delcourt and Postle [8, Corollary 3.2]). *Let $p \geq 1$ be an integer and let J be a non-null graph such that $\alpha(J') \geq v(J')/p$ for every non-null subgraph J' of J . Then $\chi(J) \leq (2 + \max\{\log(v(J)/p), 0\})p$.*

4.4 Results of Norin and Steiner

Theorem 4.15 (Norin and Steiner [36, Theorem 4]). *Let $\varepsilon \in (0, \frac{1}{2})$. There is $t_0 = t_0(\varepsilon)$ such that, for every real $t \geq t_0$, every graph G with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^{1/2-\varepsilon}$ satisfies $\chi(G) \leq (4 + \varepsilon)t$.*

Lemma 4.16 (Norin and Steiner [36, Lemma 13], the case $s = 2$). *Let G be a graph and let $V_1, \dots, V_m$ be a partition of $V(G)$ such that each $G[V_i]$ is connected and $\chi(G[V_i]) \leq 2$. If G' is obtained from G by contracting each V_i to a vertex, then $\chi(G) \leq 2\chi(G')$.*

Lemma 4.17 (Norin and Steiner [36, Corollary 17]). *Let $a, s, k \geq 0$ be real. Suppose that every graph J with $\text{had}(J) < a$ and $v(J) \leq 2ak^2$ satisfies $\chi(J) \leq s$. If H is an s -chromatic-inseparable graph with $E(H) \neq \emptyset$ and $\text{bip}(H) \leq k$, then $\text{had}(H) \geq a$.*

In applications of Lemma 4.17, we call the graphs J in its hypothesis *Gyárfás graphs*. The quantity $2ak^2$ arises as follows. The Gyárfás path argument [13], in the form of Norin and Steiner [36, Lemma 16], gives a subgraph J with $\chi(J) > s$ and $\alpha(J) \leq k(k-1) \leq k^2$, and then $v(J) \leq \chi_f(J)\alpha(J) \leq 2\text{had}(J)\alpha(J)$ by the theorem of Reed and Seymour [37] (see Norin and Steiner [36, Theorem 6]).

5 Rooted minors via massed pairs

Recall that $K_s + \ell K_2$ is the disjoint union of K_s and a matching with ℓ edges. For a graph G and $Y \subseteq V(G)$, let $\rho_G(Y)$ denote the number of edges of G with at least one end in Y . Following Thomas and Wollan [43] and Wollan [46], a pair (G, X) with $X \subseteq V(G)$ is λ -massed if

- (M1) $\rho_G(V(G) \setminus X) > \lambda |V(G) \setminus X|$, and
- (M2) every non-trivial separation (A, B) of G with $X \subseteq A$ and order at most $|X| - 1$ satisfies $\rho_G(B \setminus A) \leq \lambda |B \setminus A|$.

Wollan [46] proves his extremal theorems for rooted minors by studying a minimal counterexample. We use two of his structural results directly, rather than his final numerical bound. Let F be a graph and let $\lambda > 0$. Following Wollan, a pair (G, X) with $X \subseteq V(G)$ is (F, λ) -minimal if

- (1) $|X| \leq |V(F)|$ and (G, X) is λ -massed,
- (2) there is an injection $\pi : X \rightarrow V(F)$ such that G has no model of $F[\pi(X)]$ rooted at X according to π ,
- (3) subject to (1) and (2), $v(G)$ is as small as possible, and
- (4) subject to (1)–(3), $\rho_G(V(G) \setminus X)$ is as small as possible.

A graph J with $v(J) \geq |V(F)|$ is F -universal if, for every set $Y \subseteq V(J)$ with $|Y| = |V(F)|$ and every bijection $\pi : Y \rightarrow V(F)$, the graph J has a model of F rooted at Y according to π .

Theorem 5.1 (Wollan [46, Theorems 2.2 and 2.3]). *Let F be a graph, let $\lambda \geq |V(F)|$ be real, and let (G, X) be (F, λ) -minimal. Then the following hold.*

- (a) *If $uv \in E(G)$ with $u, v \notin X$, then u and v have at least $\lfloor \lambda \rfloor$ common neighbours. If $u \in X$, $v \notin X$, and v has exactly t' neighbours in X other than u , then u and v have at least $\lfloor \lambda \rfloor - t'$ common neighbours.*
- (b) *Some vertex of $V(G) \setminus X$ has degree at most 2λ .*
- (c) *G has no F -universal subgraph.*

These statements are on pages 163–164 of the journal version of Wollan’s paper [46] (pages 3–4 of the version on the author’s web page). Wollan’s definitions of ρ , of separations, of trivial separations and of λ -massed pairs are the ones given above. (The web version calls a separation (A, B) trivial only if $A \subseteq B$. This makes no difference, since a separation with $B \subseteq A$ has $B \setminus A = \emptyset$ and so satisfies the inequality in (M2).) Wollan’s main theorem [46, Theorem 5.1] combines Theorem 5.1 with a general argument that works for every graph F . Its consequence [46, Theorem 1.1], which is the form used by Delcourt and Postle [8, Theorem 5.9], states that every $|V(F)|$ -connected graph G with at least $(9c + 26,833|V(F)|) v(G)$ edges has a rooted model of F for every choice of roots, where c is such that every graph with at least c times as many edges as vertices has an F minor. For the graphs $F = K_s + \ell K_2$ that we need, with s much smaller than ℓ , we give a much more economical argument in Lemma 5.3 below.

Lemma 5.2. *Let $s \geq 1$ and $\ell \geq 0$ be integers, let $F := K_s + \ell K_2$, and let $\lambda \geq s + 2\ell$ be real. Suppose that for every λ -massed pair (G_0, X) with $|X| \leq s + 2\ell$ and every injection $\pi : X \rightarrow V(F)$, the graph G_0 has a model of $F[\pi(X)]$ rooted at X according to π . Then every graph G with $e(G) \geq \lambda v(G)$ and $\kappa(G) \geq 2s + 2\ell$ is (s, ℓ) -woven.*

Proof. The proof combines Wollan’s derivation of his Theorem 1.1 [46], which checks the massed conditions using connectivity, with the argument of Delcourt and Postle [8, Lemmas 5.11 and 5.13]. The latter uses the graph $K_s + \ell K_2$ and attaches each root r_i to a neighbour r'_i . Let $R = \{r_1, \dots, r_s\}$, $S = \{s_1, \dots, s_\ell\}$ and $T = \{t_1, \dots, t_\ell\}$ be as in the definition of woven. Let $I := \{i \in [s] : r_i \in S \cup T\}$. Since $\delta(G) \geq \kappa(G) \geq 2s + 2\ell$,

we can choose pairwise distinct vertices $r'_i \notin R \cup S \cup T$, for $i \in I$, with r'_i adjacent to r_i . For $i \in [s]$, let $x_i := r'_i$ if $i \in I$, and $x_i := r_i$ otherwise. Let $X := \{x_1, \dots, x_s\} \cup S \cup T$. The vertices $x_1, \dots, x_s$ are distinct and lie outside $S \cup T$. Moreover $|X| \leq s + 2\ell \leq \lambda$ and $|X| \leq \kappa(G)$.

We claim that (G, X) is λ -massed. For (M2), let (A, B) be a separation of G with $X \subseteq A$ and order at most $|X| - 1 < \kappa(G)$. Since G is $\kappa(G)$ -connected, one of $A \setminus B$ and $B \setminus A$ is empty. If $A \setminus B = \emptyset$, then $X \subseteq A \subseteq A \cap B$, which contradicts $|A \cap B| < |X|$. So $B \setminus A = \emptyset$, that is, $B \subseteq A$, and the separation is trivial. Hence there is no separation as in (M2), and (M2) holds vacuously. For (M1), since $e(G) \geq \lambda v(G)$,

$$\rho_G(V(G) \setminus X) \geq e(G) - \binom{|X|}{2} > \lambda v(G) - \lambda |X| = \lambda |V(G) \setminus X|,$$

using $\binom{|X|}{2} < \lambda |X|$.

Let $K_s + \ell K_2$ have clique vertices $q_1, \dots, q_s$ and matching edges $a_1 b_1, \dots, a_\ell b_\ell$. Define $\pi(x_i) := q_i$. If $s_j \neq t_j$, put $\pi(s_j) := a_j$ and $\pi(t_j) := b_j$. If $s_j = t_j$, put $\pi(s_j) := a_j$. Then π is a bijection from X onto a set of vertices of $K_s + \ell K_2$. Let H' be the subgraph of $K_s + \ell K_2$ induced by the image of π . By hypothesis, G contains a model $\{B_v : v \in V(H')\}$ of H' rooted at X according to π .

For each j with $s_j \neq t_j$, the sets B_{a_j} and B_{b_j} induce connected subgraphs and are joined by an edge, so their union contains an s_j - t_j path P_j . If $s_j = t_j$, let P_j be the path with the single vertex s_j . The paths P_j are pairwise disjoint and disjoint from every B_{q_i} . For $i \notin I$, let $M_i := B_{q_i}$, which contains $r_i = x_i$. For $i \in I$, let $M_i := B_{q_i} \cup \{r_i\}$. Here $r_i \in S \cup T$ lies in some set B_{a_j} or B_{b_j} , so not in any $B_{q_{i'}}$, and $G[M_i]$ is connected since $r_i r'_i \in E(G)$ and $r'_i \in B_{q_i}$. So $\{M_1, \dots, M_s\}$ is a K_s model rooted at R . A vertex in some M_i and some P_j is not in any $B_{q_{i'}}$, so it is some r_i with $i \in I$. Conversely, if $i \in I$, then r_i equals s_j or t_j for some j , and it lies on P_j . Therefore $V(\mathcal{M}) \cap V(\mathcal{P}) = R \cap (S \cup T)$. $\square$

The weaving constant. Let $M := 10^5$. In Section 9 we need that, for an integer $t \geq 3$ and $x := \lceil t/(M\sqrt{\log t}) \rceil$, every graph of sufficiently high density and moderate connectivity is $(2x, \ell)$ -woven, where $\ell := \lfloor 4.004t \rfloor$. This ℓ is an upper bound on the number of linkage pairs in each application of Lemma 4.8 in the proof of Proposition 9.1. Delcourt and Postle [8, proof of Lemma 6.4] bound the number of pairs by $10t$, which is more than needed. Delcourt and Postle also take $x = \lceil t/\sqrt{\log t} \rceil$. The factor M makes the rooted clique small, at the cost of more stages in Proposition 9.1, which affects only the size thresholds. The following lemma is the heart of this section. A minimal counterexample contains a small dense graph J , by Theorem 5.1. Since J is small and dense, a random set of at most $0.008t$ of its vertices contains a K_{2x} model, and the remaining roots and the matching pairs can then be linked around this set by Theorem 4.5.

Lemma 5.3. *Let t be an integer with $\log t \geq 1.6 \cdot 10^6$. Let $s := 2x$, where $x = \lceil t/(M\sqrt{\log t}) \rceil$, let $\ell := \lfloor 4.004t \rfloor$ and $F := K_s + \ell K_2$. Then for every $70t$ -massed pair (G, X) with $|X| \leq s + 2\ell$ and every injection $\pi : X \rightarrow V(F)$, the graph G has a model of $F[\pi(X)]$ rooted at X according to π .*

The proof follows Wollan's proof of [46, Theorem 5.1] (in particular Observation 5.2 and Claim 5.3 there). He also finds a dense subgraph in the neighbourhood of a vertex of low degree, finds a minor inside a random set of its vertices, and links the roots to this minor using the theorem of Thomas and Wollan. The differences are that we use Lemma 6.5 to obtain a highly connected dense subgraph J , and that only the clique part of F needs a minor argument, while the matching edges are routed directly by the linkage.

Proof. Let $L := \log t$, $h := s + 2\ell$, $\lambda := 70t$ and $\gamma_s := 30s\sqrt{\log s}$. Then $t > M\sqrt{L}$, so $x \geq 2$ and $s = 2x \geq 4$. Also $s \leq 2t/(M\sqrt{L}) + 2 \leq t$, so $\log s \leq L$, and hence

$$s\sqrt{\log s} \leq \left(\frac{2t}{M\sqrt{L}} + 2\right)\sqrt{L} = \frac{2t}{M} + 2\sqrt{L} \leq \frac{2.0001t}{M},$$

since $2\sqrt{L} \leq 10^{-4}t/M$. So $\gamma_s \leq 60.003t/M \leq 0.0006001t$. Also $s/t \leq 2/(M\sqrt{L}) + 2/t \leq 2 \cdot 10^{-8}$, so $h \leq 8.0081t$ and $m := s + \ell \leq 4.005t$.

Suppose the lemma fails. Then there is an (F, λ) -minimal pair (G, X) . Since $\lambda \geq h = |V(F)|$, Theorem 5.1 applies. First, no vertex v of $V(G) \setminus X$ is isolated. Indeed, otherwise $(G - v, X)$ is λ -massed: (M1) holds since ρ is unchanged and $|V(G) \setminus X|$ decreases, and a non-trivial separation (A, B) of $G - v$ as in (M2) gives the non-trivial separation $(A \cup \{v\}, B)$ of G with the same order and the same $B \setminus A$. Every rooted model in $G - v$ is one in G , so $(G - v, X)$ contradicts the minimality of $v(G)$.

By Theorem 5.1(b), some vertex $v \in V(G) \setminus X$ has degree at most 2λ . Let $D := G[\{v\} \cup N_G(v)]$, so $v(D) \leq 2\lambda + 1 \leq 141t$. We claim that $\delta(D) \geq \lambda - h$. Let $w \in N_G(v)$. If $w \notin X$, then by Theorem 5.1(a), v and w have at least λ common neighbours, all in D . If $w \in X$, then v has at most $h - 1$ neighbours in X other than w , so v and w have at least $\lambda - h + 1$ common neighbours, all in D . In both cases $\deg_D(w) \geq \lambda - h + 1$. Since v has a neighbour, the same count shows that $\deg_D(v) = \deg_G(v) \geq \lambda - h + 1$. This proves the claim. Hence $\delta(D) \geq 70t - 8.0081t \geq 61t$, and $e(D) \geq 30.5t v(D) \geq 18t v(D)$.

By Lemma 6.5 with $k = 9t$, D has a $(9t + 1)$ -connected subgraph J with $\delta(J) > 30.5t$ and $e(J) > 21.5t v(J)$. Also $v(J) \leq v(D) \leq 141t$ and $v(J) > \delta(J) > h$. We now show that J is F -universal, which contradicts Theorem 5.1(c).

Let $Y \subseteq V(J)$ with $|Y| = h$, and let $\pi : Y \rightarrow V(F)$ be a bijection. Let $K_s + \ell K_2$ have clique vertices $q_1, \dots, q_s$ and matching edges $a_1 b_1, \dots, a_\ell b_\ell$. Let $r_i := \pi^{-1}(q_i)$, $u_j := \pi^{-1}(a_j)$ and $v_j := \pi^{-1}(b_j)$.

A small clique model avoiding Y . Let $J_0 := J - Y$ and $N := v(J_0)$. Each vertex of Y lies in fewer than $v(J)$ edges, so $e(J_0) > (21.5t - h) v(J) \geq 12.5t N$. Since $N \leq 141t < 150t$, we have $e(J_0)/N > N/12$. Let $p := \lceil 12\gamma_s \rceil + 1$, so that $p \leq 0.0073t + 2 \leq 0.008t < N$. Choose a p -element subset Q of $V(J_0)$ uniformly at random. Each edge of J_0 lies in $J_0[Q]$ with probability $p(p-1)/(N(N-1))$, so

$$\frac{\mathbb{E}[e(J_0[Q])]}{p} = \frac{p-1}{N-1} \cdot \frac{e(J_0)}{N} > \frac{p-1}{N-1} \cdot \frac{N}{12} \geq \frac{p-1}{12} \geq \gamma_s.$$

So some Q satisfies $e(J_0[Q]) \geq \gamma_s |Q|$. By Theorem 4.11, $J_0[Q]$ has a K_s model $\{B_1, \dots, B_s\}$. Note that $Q \cap Y = \emptyset$ and $|Q| \leq 0.008t$.

Linking around the clique model. For each $i \in [s]$ choose $z_i \in B_i$ and a neighbour y_i of z_i in J , with $y_i \notin Q \cup Y$ and $y_1, \dots, y_s$ distinct. This is possible, since $\deg_J(z_i) > 30.5t$ and fewer than $|Q| + |Y| + s < 10t$ vertices are forbidden. The graph $J - Q$ is $(9t + 1 - p)$ -connected, and $9t + 1 - p \geq 8.99t \geq 2m$. Also $e(J - Q) \geq e(J) - p v(J) > (21.5t - 0.008t) v(J) \geq 5m v(J - Q)$. By Theorem 4.5, $J - Q$ is m -linked. The $2m$ vertices r_i, y_i ($i \in [s]$) and u_j, v_j ($j \in [\ell]$) are distinct and lie in $J - Q$. So $J - Q$ has pairwise disjoint paths P_i from r_i to y_i and P'_j from u_j to v_j . Let $B'_i := B_i \cup V(P_i)$, which induces a connected subgraph since $y_i z_i \in E(J)$. Split each path P'_j into two non-empty subpaths, one containing u_j and the other containing v_j , joined by an edge of P'_j . These sets are pairwise disjoint, because the paths avoid Q . The sets B'_i are pairwise adjacent because the sets B_i are. So they form a model of F rooted at Y according to π . Hence J is F -universal, a contradiction. $\square$

Lemma 5.4. *Let $C_{5.4} := 70$. If t is an integer with $\log t \geq 1.6 \cdot 10^6$, then every graph G with $e(G) \geq C_{5.4}t v(G)$ and $\kappa(G) \geq 9t$ is $(2x, \lfloor 4.004t \rfloor)$ -woven, where $x = \lceil t/(M\sqrt{\log t}) \rceil$ and $M = 10^5$.*

Proof. Let $s := 2x$ and $\ell := \lfloor 4.004t \rfloor$. By the proof of Lemma 5.3, $s + 2\ell \leq 8.0081t \leq 70t$ and $2s + 2\ell \leq 4 \cdot 10^{-8}t + 8.008t \leq 9t \leq \kappa(G)$. So Lemma 5.2 applies with $\lambda = 70t$, by Lemma 5.3. $\square$

For comparison, Wollan's Theorem 5.1 [46], with an optimised parameter and the bound $c \leq 4.0047t$ for the density forcing a $K_s + \ell K_2$ minor, would give $C_{5.4} \approx 4893$. Wollan's Theorem 1.1 would give $C_{5.4} \approx 9 \cdot 4.0047 + 26,833 \cdot 8.0081 \approx 214,900$. The difference is that the general argument must handle an arbitrary graph F , whereas for $K_s + \ell K_2$ with s small only the clique needs a minor-type argument, and the matching is handled by Theorem 4.5.

6 A refined form of Lin's theorem

The chromatic-inseparable case (Section 9) needs to find small highly connected subgraphs in dense graphs with no K_t minor. Delcourt and Postle use Theorem 4.12. Lin [24] proved the following improvement of the order of the pieces. Lin states it with an unspecified constant, and his proof gives the value $\max\{30300, 32C_{6.3}^2\}$, where $C_{6.3}$ is the constant of the Norin–Postle bound. The explicit form below is obtained by substituting the value $C_{6.3} = 6400$ of Norin and Postle [34]. This is the constant $C_{6.3}$ of Theorem 6.3 below.

Theorem 6.1 (Lin [24, Theorem 1.4]). *Let $C_{6.1} := \max\{30300, 32 \cdot 6400^2\}$. For all integers $t \geq 3$ and $k \geq t$, every non-null graph G with no K_t minor and $e(G) \geq C_{6.1}k v(G)$ has a non-null k -connected subgraph with at most $C_{6.1}^2 t \log t$ vertices.*

The density constant $C_{6.1} \approx 1.31 \cdot 10^9$ is quadratic in the Norin–Postle constant. It is much larger than the constant $C_{4.12} \approx 3.07 \cdot 10^6$ of Theorem 4.12. In our application the density constant determines $C_{9.1}$, which enters the final bound only through the term $2C_{9.1}$ (from the chromatic threshold), while the order of the pieces affects only the size thresholds. We therefore rerun Lin's argument with different parameters. We also extract a subgraph that is dense, not only highly connected. This is what the chromatic-inseparable case really needs (see Lemma 5.4). The result is the following.

Theorem 6.2. *Let $\omega_{6.2} := 0.0044$ and $\tau_{6.2} := 4.56 \cdot 10^{18}$. Let $t \geq 10^{13}$ and $d \geq 1.8 \cdot 10^4 t$ be integers, and let k be an integer with $1 \leq k \leq \omega_{6.2}d/2$. Every non-null graph G with no K_t minor and $e(G) \geq d v(G)$ has a $(k+1)$ -connected subgraph H with at most $\tau_{6.2} t \log t$ vertices and with $e(H) > (\omega_{6.2}d - k) v(H)$.*

The improvement over Lin's theorem has four sources. First, when d is large compared with t , the linear term in the Norin–Postle bound is small compared with $d v(G)$. Second, the bound Q below, on the weight of the contracted subgraphs, is taken very large. This makes the other term of the Norin–Postle bound small as well, at the cost of a larger constant $\tau_{6.2}$. Third, the parameters of the dense-subset lemma are balanced against each other. Fourth, in the last step we extract a dense subgraph using a simple averaging lemma (Lemma 6.6), and then we use the proof of Mader's theorem [29] to obtain a highly connected subgraph whose density is close to that of the dense subgraph (Lemma 6.5). Besides Theorem 4.11, the proof uses four further tools.

Theorem 6.3 (Norin and Postle [34, Theorem 3.2], see also [8, Theorem 4.1]). *Let $C_{6.3} := 6400$ and let $t \geq 3$ be an integer. Let B be a bipartite graph with no K_t minor and with bipartition (V_1, V_2) . Then*

$$e(B) \leq C_{6.3} t \sqrt{\log t} \sqrt{|V_1||V_2|} + (t-2)(|V_1| + |V_2|).$$

Lemma 6.4 (Delcourt and Postle [8, Lemma 4.2], see also [24, Lemma 2.2]). *Let $r > 2$ and $\delta_0 > 0$ be real. Let G be a graph and $S \subseteq V(G)$. If*

$$(r - 2)e(G[S]) > (r - 1)\delta_0|S| + e_G(S, V(G) \setminus S),$$

then there is a non-empty set $S' \subseteq S$ such that $\delta(G[S']) \geq \delta_0$ and $|N_G(v) \cap S'| \geq \deg_G(v)/r$ for every $v \in S'$.

Lemma 6.5 (Mader [29], see [9, Theorem 1.4.3]). *Let $k \geq 1$ be an integer, and let G be a non-null graph with $e(G) \geq 2kv(G)$. Let $\gamma := e(G)/v(G)$. Let $\gamma' := \frac{1}{2}(\gamma + \sqrt{\gamma^2 - 2\gamma k})$. Then G has a $(k + 1)$ -connected subgraph H with $\delta(H) > \gamma$, $v(H) > 2\gamma'$ and $e(H) > \gamma'v(H) \geq (\gamma - k)v(H)$.*

Proof. The bounds $\delta(H) > \gamma$ and $e(H) > (\gamma - k)v(H)$ are in the statement and proof of [9, Theorem 1.4.3]; the bounds $v(H) > 2\gamma'$ and $e(H) > \gamma'v(H)$ are a slight sharpening from the same proof. We follow the proof of [9, Theorem 1.4.3], which records the bounds on $\delta(H)$ and $e(H)$ along the way. Note that $\gamma \geq 2k$. Call a subgraph G' of G *good* if $v(G') \geq 2k$ and $e(G') > \gamma(v(G') - k)$. The graph G itself is good, since $e(G) = \gamma v(G)$ and $v(G) > 2\gamma \geq 4k$ (the average degree 2γ is less than $v(G)$). Let H be a good subgraph with as few vertices as possible. No good graph has exactly $2k$ vertices, since it would have more than $\gamma k \geq 2k^2 > \binom{2k}{2}$ edges. So $v(H) > 2k$. If H had a vertex v with $\deg_H(v) \leq \gamma$, then $H - v$ would have $v(H) - 1 \geq 2k$ vertices and $e(H - v) > \gamma(v(H) - k) - \gamma = \gamma(v(H - v) - k)$ edges. So $H - v$ would be good, contradicting the choice of H . Hence $\delta(H) > \gamma$. In particular $v(H) > \gamma$. Let $n := v(H)$ and $\rho := \sqrt{\gamma^2 - 2\gamma k}$, which is real since $\gamma \geq 2k$. Since H is good and $e(H) \leq n(n - 1)/2 < n^2/2$, we have $n^2/2 > \gamma(n - k)$, that is, $n^2 - 2\gamma n + 2\gamma k > 0$. The roots of this quadratic are $\gamma \pm \rho$, and $n > \gamma \geq \gamma - \rho$, so $n > \gamma + \rho = 2\gamma'$. Hence

$$\frac{e(H)}{n} > \gamma \left(1 - \frac{k}{n}\right) > \gamma - \frac{\gamma k}{\gamma + \rho} = \gamma - \frac{\gamma - \rho}{2} = \gamma',$$

using $(\gamma + \rho)(\gamma - \rho) = 2\gamma k$. Finally $\gamma' \geq \gamma - k$, since $\rho \geq \gamma - 2k \geq 0$ is equivalent to $2\gamma k \geq 4k^2$.

It remains to show that H is $(k + 1)$ -connected. We have $v(H) > 2k \geq k + 1$. Suppose that H has a separation (U_1, U_2) with $|U_1 \cap U_2| \leq k$ and both $U_1 \setminus U_2$ and $U_2 \setminus U_1$ non-empty. Let $H_i := H[U_i]$. A vertex of $U_1 \setminus U_2$ has all its neighbours in U_1 , so $v(H_1) > \delta(H) > \gamma \geq 2k$. Similarly $v(H_2) > 2k$. Each H_i has fewer vertices than H , so it is not good, and hence $e(H_i) \leq \gamma(v(H_i) - k)$. Every edge of H lies in H_1 or H_2 , and $v(H_1) + v(H_2) \leq v(H) + k$. Therefore

$$e(H) \leq e(H_1) + e(H_2) \leq \gamma(v(H_1) + v(H_2) - 2k) \leq \gamma(v(H) - k),$$

contradicting the fact that H is good. So H is $(k + 1)$ -connected. $\square$

Lemma 6.6. *Let F be a graph, let $A \subseteq V(F)$ be non-empty, and let $D \geq 0$. Let $\gamma := D|A|/(v(F) + |A|)$. If $\deg_F(v) \geq D$ for every $v \in A$, then $e(F) \geq \gamma v(F)$ or $e(F[A]) \geq \gamma|A|$.*

Proof. Each edge of $F[A]$ contributes 2 to $\sum_{v \in A} \deg_F(v)$, and each other edge contributes at most 1. So $D|A| \leq \sum_{v \in A} \deg_F(v) \leq 2e(F[A]) + (e(F) - e(F[A])) = e(F[A]) + e(F)$. If $e(F) < \gamma v(F)$ and $e(F[A]) < \gamma|A|$, then $e(F) + e(F[A]) < \gamma(v(F) + |A|) = D|A|$, a contradiction. $\square$

Lin [24, Theorem 2.4] and Delcourt and Postle [8, Lemma 4.4] use only the connectivity in Lemma 6.5. The average degree $4k$ needed for a $(k + 1)$ -connected subgraph has been reduced to about $(3 + \frac{1}{6})k$ by Bernshteyn and Kostochka [3]. Carmesin [7] proved a

sharp form of Mader's version, in which the subgraph has more than $2k$ vertices. These improvements do not help us, because we need the density bound in Lemma 6.5, and the connectivity we need is much smaller than the density.

We use the following notation. For a graph J and $A \subseteq V(J)$, let $w_J(A) := \sum_{v \in A} \deg_J(v)$. Contraction is as defined in Section 4.

Proof of Theorem 6.2. The proof follows Lin [24, Section 3], which in turn follows the proof of a theorem of Delcourt and Postle [8, Theorem 2.3]. Claims 1–3 below are Lin's claims [24, Claims 3.1–3.3] with our parameters. Let $n := v(G)$. Then $d \geq 1.8 \cdot 10^{17}$. By deleting edges, we may assume that $e(G) = dn$, since d and n are integers. A subgraph of the new graph is a subgraph of the original graph, so this is no loss. Then

$$w_G(V(G)) = 2e(G) = 2dn \leq 2.002dn.$$

Below we use only the weaker bound $w_G(V(G)) \leq 2.002dn$. The extra slack (2.002 rather than 2) is harmless, and we keep it for convenience. Since G has no K_t minor and $e(G) \geq dn$, Theorem 4.11 gives $d < 30t\sqrt{\log t}$. Define the parameters

$$\Xi := 2 \cdot 10^{15}, \quad \varepsilon := 0.083, \quad \theta := 0.003, \quad \eta' := 0.02, \quad r := 2.92, \quad \delta_0 := 0.37d,$$

and let

$$Q := \frac{\Xi C_{6.3}^2 t^2 \log t}{d}.$$

Step 1. Contracting heavy subgraphs: Choose pairwise vertex-disjoint non-null connected subgraphs $H_1, \dots, H_m$ of G with the following two properties, with m as large as possible. First, $w_G(V(H_i)) \geq Q$ for each i . Second, the graph G' obtained by contracting each H_i to a new vertex x_i satisfies

$$e(G) - e(G') \leq \varepsilon \sum_{i=1}^m w_G(V(H_i)).$$

The empty family has both properties, so such a family exists. The graph G' is a minor of G , so it has no K_t minor. Let $X := \{x_1, \dots, x_m\}$ and $O := V(G') \setminus X$. Since the sets $V(H_i)$ are disjoint, $|X|Q \leq w_G(V(G)) \leq 2.002dn$. Hence

$$|X| \leq \frac{2.002dn}{Q} = \frac{2.002d^2n}{\Xi C_{6.3}^2 t^2 \log t}.$$

Also $e(G') \geq e(G) - 2.002\varepsilon dn \geq (1 - 2.002\varepsilon)dn$.

Claim 1. There is no non-empty set $A \subseteq O$ such that $G'[A]$ is connected, $w_{G'}(A) \geq Q$ and $e(G') - e(G'/A) \leq \varepsilon w_{G'}(A)$. In particular, $\deg_{G'}(v) < Q$ for every $v \in O$.

Proof of Claim 1. The contractions forming G' do not change adjacencies within O . So $G'[O] = G[O]$, and $\deg_{G'}(v) \leq \deg_G(v)$ for $v \in O$. Suppose that A exists. Then $w_G(A) \geq w_{G'}(A) \geq Q$. Moreover G'/A is obtained from G by contracting $H_1, \dots, H_m$ and $G[A]$. Since

$$e(G) - e(G'/A) = (e(G) - e(G')) + (e(G') - e(G'/A)) \leq \varepsilon \sum_{i=1}^m w_G(V(H_i)) + \varepsilon w_G(A),$$

the family $H_1, \dots, H_m, G[A]$ contradicts the maximality of m . For the second statement, take $A = \{v\}$, for which $e(G') - e(G'/A) = 0$. $\diamond$

Step 2. Removing the vertices attached to X : Let

$$Y := \{v \in O : |N_{G'}(v) \cap X| \geq \theta \deg_{G'}(v)\},$$

and let $T := X \cup Y$ and $S := V(G') \setminus T$.

Claim 2. $e(G'[T]) + e_{G'}(S, T) \leq \eta' dn$.

Proof of Claim 2. Apply Theorem 6.3 to $G'(X, O)$. Since $|O| \leq n$ and $|X| + |O| \leq n$, the bound on $|X|$ gives

$$e_{G'}(X, O) \leq C_{6.3} t \sqrt{\log t} \sqrt{|X|n} + tn \leq \sqrt{\frac{2.002}{\Xi}} dn + \frac{t}{d} dn \leq \nu dn,$$

where $\nu := \sqrt{2.002/\Xi} + 1/(1.8 \cdot 10^4) < 5.559 \cdot 10^{-5}$. By the definition of Y ,

$$\theta \sum_{v \in Y} \deg_{G'}(v) \leq e_{G'}(X, Y) \leq e_{G'}(X, O),$$

so $\sum_{v \in Y} \deg_{G'}(v) \leq (\nu/\theta) dn$. If $X \neq \emptyset$, then Theorem 4.11 and the bound on $|X|$ give

$$e(G'[X]) < 30t \sqrt{\log t} |X| \leq \frac{60.06}{\Xi C_{6.3}^2} \cdot \frac{d}{t \sqrt{\log t}} \cdot dn < 10^{-19} dn,$$

using $d < 30t \sqrt{\log t}$. This bound is trivial if $X = \emptyset$. Every edge counted on the left of Claim 2 lies in $G'[X]$, joins X to O , or has an end in Y . Hence

$$\begin{aligned} e(G'[T]) + e_{G'}(S, T) &\leq e(G'[X]) + e_{G'}(X, O) + \sum_{v \in Y} \deg_{G'}(v) \\ &\leq (10^{-19} + \nu(1 + 1/\theta)) dn < 0.01859 dn < \eta' dn. \end{aligned}$$

This proves Claim 2. $\diamond$

Step 3. A dense subset: By Claim 2,

$$e(G'[S]) = e(G') - e(G'[T]) - e_{G'}(S, T) \geq (1 - 2.002\varepsilon - \eta') dn = 0.813834 dn.$$

Since $|S| \leq n$, Claim 2 also gives

$$\begin{aligned} (r-1) \delta_0 |S| + e_{G'}(S, V(G') \setminus S) &\leq (1.92 \cdot 0.37 + \eta') dn < 0.7305 dn \\ &< 0.92 \cdot 0.813834 dn \leq (r-2) e(G'[S]). \end{aligned}$$

By Lemma 6.4, there is a non-empty set $S' \subseteq S$ such that $\delta(G'[S']) \geq \delta_0$ and $|N_{G'}(v) \cap S'| \geq \deg_{G'}(v)/r$ for every $v \in S'$.

Step 4. The components of $G'[S']$ are heavy: Since $d \geq 1.8 \cdot 10^4 t$,

$$\frac{Q}{\delta_0^2} = \frac{\Xi C_{6.3}^2 t^2 \log t}{0.37^2 d^3} \leq \frac{\Xi C_{6.3}^2}{0.37^2 \cdot (1.8 \cdot 10^4)^3} \cdot \frac{\log t}{t} < 1.0261 \cdot 10^{11} \frac{\log t}{t} < 1,$$

because $t \geq 10^{13}$. Let K be a component of $G'[S']$. Then $v(K) > \delta_0$, and so $w_{G'}(V(K)) \geq \delta_0 v(K) > \delta_0^2 > Q$.

Step 5. A maximal light set: Choose a non-empty set $U \subseteq S'$ such that $G'[U]$ is connected, $w_{G'}(U) < Q$ and $e(G') - e(G'/U) \leq \varepsilon w_{G'}(U)$, with $|U|$ as large as possible. Every singleton in S' has these properties, by Claim 1. Let K_U be the component of $G'[S']$ containing U . By Step 4, $w_{G'}(V(K_U)) > Q > w_{G'}(U)$. So $U \neq V(K_U)$, and some vertex of $S' \setminus U$ has a neighbour in U .

Claim 3. Let $z \in S' \setminus U$ have a neighbour in U . Then $e(G') - e(G'/(U \cup \{z\})) > \varepsilon w_{G'}(U \cup \{z\})$.

Proof of Claim 3. Suppose not. The graph $G'[U \cup \{z\}]$ is connected. If $w_{G'}(U \cup \{z\}) < Q$, then $U \cup \{z\}$ contradicts the maximality of $|U|$. Otherwise $w_{G'}(U \cup \{z\}) \geq Q$, and since $U \cup \{z\} \subseteq S' \subseteq O$, this contradicts Claim 1. $\diamond$

Step 6. A dense neighbourhood: Let $G'' := G'/U$, and let x be the vertex obtained by contracting U . Let $R := N_{G''}(x) \setminus X$ and $R' := N_{G''}(x) \cap (S' \setminus U)$. Then $R' \subseteq R$, since $S' \cap X = \emptyset$. Also $|R| \leq w_{G'}(U)$. Counting the edges lost when U is contracted gives

$$\begin{aligned} \sum_{u \in U} |N_{G'}(u) \cap S'| - |R'| &= 2e(G'[U]) + \sum_{\substack{y \in S' \setminus U \\ N_{G'}(y) \cap U \neq \emptyset}} (|N_{G'}(y) \cap U| - 1) \\ &\leq 2(e(G') - e(G'/U)) \leq 2\varepsilon w_{G'}(U). \end{aligned}$$

By the choice of S' , the first sum is at least $w_{G'}(U)/r$. Hence

$$|R'| \geq \left(\frac{1}{r} - 2\varepsilon\right)w_{G'}(U) \geq 0.1764 w_{G'}(U) \geq 0.1764 |R|.$$

By Step 5, some vertex of $S' \setminus U$ has a neighbour in U , and this vertex lies in R' . So R' , and hence R , is non-empty. Let $z \in R'$ be arbitrary. Contracting U and then the edge xz has the same effect as contracting $U \cup \{z\}$ in G' . By Claim 3 and the choice of U ,

$$\begin{aligned} e(G'') - e(G''/xz) &= (e(G') - e(G'/(U \cup \{z\}))) - (e(G') - e(G'/U)) \\ &> \varepsilon(w_{G'}(U \cup \{z\}) - w_{G'}(U)) = \varepsilon \deg_{G'}(z). \end{aligned}$$

Contracting an edge of a simple graph deletes that edge and one further edge for each common neighbour of its ends. So x and z have more than $\varepsilon \deg_{G'}(z) - 1$ common neighbours in G'' . Since $z \in S$, we have $z \notin Y$, and so $|N_{G'}(z) \cap X| < \theta \deg_{G'}(z)$. Contracting U does not change the neighbours of z in X . Hence more than $(\varepsilon - \theta) \deg_{G'}(z) - 1$ common neighbours of x and z lie outside X , and all of them lie in R . Since $\deg_{G'}(z) \geq \delta_0 = 0.37d$ and $d \geq 1.8 \cdot 10^{17}$,

$$|N_{G''}(z) \cap R| > 0.08 \cdot 0.37d - 1 \geq 0.02959d.$$

This holds for every $z \in R'$. No vertex of R is involved in the contractions forming G'' , so $G''[R] = G[R]$. Apply Lemma 6.6 to $F := G[R]$ and $A := R'$, with $D := 0.02959d$. Since $|R'| \geq 0.1764 |R|$ and $\frac{D|A|}{v(F)+|A|}$ is increasing in $|A|$, one of $G[R]$ and $G[R']$ has at least

$$\frac{0.02959 \cdot 0.1764}{1 + 0.1764} d > 0.0044d = \omega_{6.2}d$$

times as many edges as vertices. Let R^* be R or R' accordingly, so that $R^* \subseteq R$ and $e(G[R^*]) \geq \omega_{6.2}d |R^*|$.

Step 7. Conclusion: Let $\gamma := e(G[R^*])/|R^*|$. By Step 6, $\gamma \geq \omega_{6.2}d \geq 2k$. By Lemma 6.5, $G[R^*]$ has a $(k+1)$ -connected subgraph H with $e(H) > (\gamma - k)v(H) \geq (\omega_{6.2}d - k)v(H)$. Finally, since $d \geq 1.8 \cdot 10^4 t$,

$$v(H) \leq |R| \leq w_{G'}(U) < Q = \frac{\Xi C_{6.3}^2 t}{d} \cdot t \log t \leq \frac{\Xi C_{6.3}^2}{1.8 \cdot 10^4} t \log t < \tau_{6.2} t \log t.$$

Here $\Xi C_{6.3}^2 / (1.8 \cdot 10^4) < 4.552 \cdot 10^{18}$. $\square$

Since $d < 30t\sqrt{\log t}$ and $d \geq 1.8 \cdot 10^4 t$, Theorem 6.2 is vacuous unless $\log t > 3.6 \cdot 10^5$. In particular, the hypotheses can be satisfied only if $t \geq 10^{13}$, and this condition is included only to make Step 4 transparent. Similarly, in the application in Section 9 the density hypothesis can hold only for enormous t . Otherwise Theorem 4.11 already shows that no subgraph is that dense. The statement is nevertheless needed, since our bounds must hold for all t .

The proof of Lin [24] is the case $d = C_{6.1}k$, with $Q = 8C_{6.3}^2 d \log t (t/k)^2$, $\varepsilon = \frac{1}{100}$, $\theta = \frac{1}{200}$, $\eta' = \frac{1}{100}$, $r = 3$ and $\delta_0 = \frac{2}{5}d$, and it uses only the connectivity in Lemma 6.5. With this choice of Q , the two terms of the Norin–Postle bound in Claim 2 are at most $kn/2$ and at most kn , so together at most $\frac{3}{2}kn$. This forces d/k to be large compared with $1/(\eta'\theta)$, and gives Lin's term 30300. Lin's other term $32C_{6.3}^2$ comes from the analogue of Step 4, where Lin needs $Q < \delta_0^2$ using only $\log t \leq k/2$. In our proof, the two terms of the Norin–Postle bound together give $e_{G'}(X, O) \leq \nu dn$ with $\nu < 5.559 \cdot 10^{-5}$, because Ξ is very large and $d \geq 1.8 \cdot 10^4 t$. After the factor $1 + 1/\theta$ coming from Y , the total in Claim 2 is less than $0.0186 dn$.

7 An explicit chromatic-separable reduction

This section makes explicit a lemma and a theorem of Norin and Steiner [36, Lemma B.12 and Theorem 15]. These are adaptations of results of Delcourt and Postle [8, Theorem 7.1 and Theorem 1.6]. We first construct a linked subgraph of bounded chromatic number, which serves as a hub.

Lemma 7.1. *Let $m \geq 1$ be an integer and let $q := \lceil 25m/2 \rceil + 1$. Every graph F with $\chi(F) \geq q$ has an m -linked induced subgraph H_0 with $v(H_0) > 10m$ and $\chi(H_0) \leq q$.*

Proof. The idea of first trimming the chromatic number to an exact value is taken from the proof of a lemma of Norin and Steiner [36, Lemma B.12]. Deleting a vertex lowers the chromatic number by at most 1. So F has an induced subgraph F_0 with $\chi(F_0) = q$ such that $\chi(F_0 - v) < q$ for every vertex v of F_0 . So F_0 is vertex-critical, and hence $\delta(F_0) \geq q - 1$, a classical property of critical graphs (see Dirac [10]). Thus $\delta(F_0) \geq 25m/2$, so $\gamma := e(F_0)/v(F_0) \geq 25m/4 \geq 4m$. Apply Lemma 6.5 to F_0 with $2m$ in place of k . The function $\gamma \mapsto \frac{1}{2}(\gamma + \sqrt{\gamma^2 - 4m\gamma})$ is increasing for $\gamma \geq 4m$, and equals $5m$ at $\gamma = 25m/4$. So F_0 has a $(2m + 1)$ -connected subgraph H with $e(H) > 5m v(H)$ and $v(H) > 10m$. By Theorem 4.5, H is m -linked. Let $H_0 := F[V(H)]$. Then H_0 contains H as a spanning subgraph, so it is m -linked, and $H_0 = F_0[V(H)]$, so $\chi(H_0) \leq \chi(F_0) = q$. $\square$

The use of density here avoids the factor 10 in the connectivity form of Theorem 4.5. Lemma 7.1 uses the sharper density bound in Lemma 6.5; the weaker bound $e(H) > (\gamma - k)v(H)$ would give only $4.25m$ in place of $5m$.

Lemma 7.2. *Let $\eta \in (0, 1]$, $a_0 \geq 4 + 5/\eta$ and $\Phi \geq 0$ be real. Let*

$$\Lambda_{7.2}^E := (1 + \eta)(517.5 + 84(1 + \Phi)), \quad \Lambda_{7.2}^U := (1 + \eta)(897.5 + \frac{230}{3}(1 + \Phi)),$$

and let $\Lambda := \min\{\Lambda_{7.2}^E, \Lambda_{7.2}^U\}$. Let $a \geq 1$ be an integer, and let G be a $21a$ -connected graph that satisfies $(\mathcal{H}_{\Phi, a_0})$ and

$$\chi(G) \geq D\left(\frac{46}{3}a_0 + 4\right) + 1 + \Lambda a.$$

Then G is $(a, 3a)$ -woven if $\Lambda = \Lambda_{7.2}^E$, and $(a, \lceil 10a/3 \rceil)$ -woven otherwise. In particular, G has a K_a minor.

Proof. The proof follows the proof of a theorem of Delcourt and Postle [8, Theorem 7.1], and its adaptation by Norin and Steiner [36, Lemma B.12]. Steps 2 and 5–9 follow these proofs, with hypothesis $(\mathcal{H}_{\Phi, a_0})$ in place of their bound on inseparability, and Steps 1, 3 and 4 fill in details of these proofs. The case $a \leq a_0$ in Step 2 adapts the base case of Norin and Steiner [36, Lemma B.12]. The three applications of Lemma 4.8 in Step 9 are a claim of Delcourt and Postle [8, Claim 7.1.1]. In more detail, Step 1 gives the reduction to pairwise disjoint R , S and T , which Delcourt and Postle [8, proof of Theorem 7.1] assert using the minimum degree; replacing a root by a neighbour is as in their proof of [8, Lemma 5.13]. The role of the hub in Step 3 is as in their proof, and the new ingredient is its construction by Lemma 7.1, in place of the results used by Norin and Steiner [36, Theorems B.1 and B.3]. Step 4 gives the details of their application of Menger’s theorem, including the colouring of induced paths with two colours each. In Step 6, Delcourt and Postle apply their Lemma 5.3 directly to paths starting in $V(G_3)$; the identification trick from their proof of [8, Lemma 6.4] makes this application valid. The remaining new ingredient is in Step 9. In Step 9 the branch sets are distributed unevenly among the three subgraphs: the subgraph that costs only one separation receives more of them. This is why we prove $(a, \lceil 10a/3 \rceil)$ -wovenness rather than $(a, 3a)$ -wovenness, and it reduces the coefficient of Φ from 84 to $\frac{230}{3}$, at the cost of increasing the constant term from 517.5 to 897.5. We call this the *uneven scheme*, and the even distribution of Delcourt and Postle the *even scheme*. The even scheme is better when $\Phi < 559/11 \approx 50.82$, which is the case in the bootstrapping of Section 8. The uneven scheme is used in the final application (Lemma 10.9), where the role of $1 + \Phi$ is played by $78 + 2c_{10.5} \approx 2.5 \cdot 10^5$. The scheme is determined by Φ , so it is the same at every level of the induction below. We give the proof for the uneven scheme ($\Lambda = \Lambda_{7.2}^U$), and then list the changes for the even scheme.

We proceed by induction on a . Every subgraph extracted below by Corollary 4.3 is induced. Hence every graph to which the induction hypothesis is applied is an induced subgraph of G and satisfies $(\mathcal{H}_{\Phi, a_0})$. Throughout, let $\ell := \lceil 10a/3 \rceil$ and $h := 2a + 4\ell$. Then

$$\ell \leq \frac{10}{3}a + 1, \quad a + \ell \leq \frac{13}{3}a + 1, \quad a + 2\ell \leq \frac{23}{3}a + 2, \quad h \leq \frac{46}{3}a + 4 \leq \frac{46}{3}(a + 1).$$

Step 1. Reduction to disjoint terminals: Let $R = \{r_1, \dots, r_a\}$, $S = \{s_1, \dots, s_\ell\}$ and $T = \{t_1, \dots, t_\ell\}$ be as in the definition of woven. We modify the terminals, using that $\delta(G) \geq 21a$. First, for each i with $s_i = t_i$, choose a neighbour t_i^* of s_i not in $R \cup S \cup T$ and distinct from all previously chosen vertices, and replace t_i by t_i^* . Second, for each $r_i \in S \cup T$ (with T the original set), choose a neighbour r_i^* of r_i not in $R \cup S \cup T$ and distinct from all previously chosen vertices, and replace r_i by r_i^* . At most $a + \ell$ new vertices are chosen. At each step fewer than $(a + 2\ell) + (a + \ell) \leq 12a + 3 \leq 21a$ vertices are forbidden, so the choices are possible. Call the new sets R^*, S, T^* . They are pairwise disjoint.

Suppose that $\mathcal{M}^* = \{M_i^*\}$ is a K_a model rooted at R^* and $\mathcal{P}^*$ is an $\{(s_i, t_i^*)\}$ -linkage with $V(\mathcal{M}^*) \cap V(\mathcal{P}^*) = \emptyset$. For each i with $s_i = t_i$, replace the path of $\mathcal{P}^*$ from s_i to t_i^* by the one-vertex path s_i . For each i with $r_i \neq r_i^*$, add r_i to M_i^* . This keeps $G[M_i^*]$ connected, since $r_i r_i^* \in E(G)$. The result is a K_a model rooted at R and an $\{(s_i, t_i)\}$ -linkage. A vertex lies in both only if it is some $r_i \in S \cup T$, which is an end of a path. So the intersection is exactly $R \cap (S \cup T)$. Hence it suffices to treat pairwise disjoint sets R, S, T , which we assume from now on.

Let $Z := R \cup S \cup T$, so $|Z| = a + 2\ell$. Index Z as $z_i := s_i$ ($1 \leq i \leq \ell$), $z_{\ell+i} := r_i$ ($1 \leq i \leq a$) and $z_{a+\ell+i} := t_i$ ($1 \leq i \leq \ell$).

Step 2. The base cases: Suppose G has a K_h minor. Since $h = 2|Z|$ and $\kappa(G) \geq 21a \geq |Z|$, Lemma 4.9 shows that G is $(|Z|, 0)$ -woven, so G has a $K_{|Z|}$ model $\{B_z : z \in Z\}$ rooted at Z . The sets B_{r_i} form a K_a model rooted at R . For each $i \leq \ell$, the union $B_{s_i} \cup B_{t_i}$ contains an s_i – t_i path. These paths are pairwise disjoint and disjoint from the sets B_{r_i} .

So G is (a, ℓ) -woven. We may therefore assume that G has no K_h minor. If $a \leq a_0$, then $h \leq \frac{46}{3}a_0 + 4$, and so $\chi(G) \leq D(h) \leq D(\frac{46}{3}a_0 + 4)$, contrary to the hypothesis. Hence $a > a_0 \geq 4 + 5/\eta \geq 9$.

Step 3. The hub: Let $G_1 := G - Z$. Then $\kappa(G_1) \geq 21a - |Z| \geq 2a$ and $\chi(G_1) \geq \chi(G) - |Z| \geq \lceil 12.5(a + \ell) \rceil + 1$. By Lemma 7.1 with $a + \ell$ in place of m , G_1 has an $(a + \ell)$ -linked induced subgraph H_0 , which we call the *hub*, with $v(H_0) > 10(a + \ell)$ and $\chi(H_0) \leq \lceil 12.5(a + \ell) \rceil + 1 \leq 12.5(a + \ell) + 2$.

Step 4. Paths to the hub: Let G^+ be obtained from G by adding, for each $z \in Z$, a new vertex z' adjacent to z and to every neighbour of z . Adding a vertex with at least $21a$ neighbours preserves $21a$ -connectivity, so G^+ is $21a$ -connected, and hence $2|Z|$ -connected since $2|Z| \leq \frac{46}{3}a + 4 \leq 21a$. Since $v(H_0) > 10(a + \ell) \geq 2|Z|$, Menger's theorem (Theorem 4.4, with $2|Z|$ in place of k) gives $2|Z|$ pairwise vertex-disjoint $(Z \cup Z') - V(H_0)$ paths in G^+ , where $Z' = \{z' : z \in Z\}$. A path starting at z' does not contain z , so its second vertex is a neighbour of z . Replacing z' by z therefore gives a path in G . We obtain a family $\mathcal{P}_1$ of $2|Z|$ $Z - V(H_0)$ paths in G , pairwise disjoint outside Z , in which every $z \in Z$ is an end of exactly two paths. Replace each path by a shortest path between its ends in the subgraph induced by its own vertex set. This preserves the properties above, and each path becomes an induced path. The subgraph of G induced by $V(\mathcal{P}_1) \setminus Z$ can therefore be coloured by giving the vertices of each path two colours of its own, so it has chromatic number at most $4|Z|$.

Step 5. The next high-chromatic subgraph: Let $G_2 := G - (Z \cup V(\mathcal{P}_1) \cup V(H_0))$. Then

$$\chi(G_2) \geq \chi(G) - 5|Z| - 12.5(a + \ell) - 2 \geq \chi(G) - (92.5a + 24.5),$$

which is at least $\lceil 3.0625(21a - 1) \rceil$. By Corollary 4.3, G_2 has a $21a$ -connected induced subgraph G_3 with $\chi(G_3) \geq \chi(G_2) - 42a$.

Step 6. Reduction to G_3 : We show that if G_3 is $(a, 0)$ -woven, then G is (a, ℓ) -woven. Since G_1 is $2a$ -connected, Menger's theorem (Theorem 4.4) gives a family $\mathcal{P}_2$ of $2a$ pairwise disjoint $V(G_3) - V(H_0)$ paths in G_1 . Pair these paths arbitrarily into a pairs. Let $G^\times$ be the graph obtained from $G[V(\mathcal{P}_1) \cup V(\mathcal{P}_2) \cup V(H_0)]$ by identifying the first vertices of the two paths of each pair to a new vertex c_i (as defined in Section 4), and let $C := \{c_1, \dots, c_a\}$. In $G^\times$ the hypotheses of Lemma 4.7 hold with $A_1 = Z$, $A_2 = C$ and $B = V(H_0)$:

- $\mathcal{P}_1$ consists of $Z - V(H_0)$ paths avoiding C , because they avoid $G_3 \subseteq G_2$.
- $\mathcal{P}_2$ consists of $C - V(H_0)$ paths avoiding Z , because they lie in G_1 .
- Each c_i lies in exactly two paths of $\mathcal{P}_2$, and the paths of $\mathcal{P}_2$ are pairwise disjoint outside C .

Let $N := |Z| + a = 2(a + \ell)$. Lemma 4.7 gives N pairwise disjoint $(Z \cup C) - V(H_0)$ paths in $G^\times$. Each of them has exactly one vertex in the set $Z \cup C$ of size N , so exactly one starts at each vertex of $Z \cup C$. Undoing the identification, we obtain pairwise disjoint paths $P_1, \dots, P_N$ in G , where P_i starts at z_i for $i \leq |Z|$, and $P_{|Z|+i}$ starts at a vertex $v_{|Z|+i} \in V(G_3)$ for $i \leq a$. Let $u_i \in V(H_0)$ be the last vertex of P_i . Each P_i meets $V(H_0)$ only in u_i . It meets $Z \cup V(G_3)$ only in its first vertex, because the paths of $\mathcal{P}_2$ meet $V(G_3)$ only in their first vertices and the paths of $\mathcal{P}_1$ avoid $V(G_3)$.

Since G_3 is $(a, 0)$ -woven, it has a K_a model $\{M'_1, \dots, M'_a\}$ with $v_{|Z|+i} \in M'_i$. Since H_0 is $(a + \ell)$ -linked, it contains pairwise disjoint paths $Q_1, \dots, Q_{a+\ell}$ with Q_i joining u_i to $u_{a+\ell+i}$. For $i \leq a$, let

$$M_i := V(P_{\ell+i}) \cup V(Q_{\ell+i}) \cup V(P_{|Z|+i}) \cup M'_i.$$

Here $Q_{\ell+i}$ joins $u_{\ell+i}$ to $u_{a+2\ell+i} = u_{|Z|+i}$. Then $G[M_i]$ is connected and contains $z_{\ell+i} = r_i$, the sets M_i are pairwise disjoint, and any two are adjacent because the M'_i are. So $\{M_i\}$ is

a K_a model rooted at R . For $i \leq \ell$, the union $P_i \cup Q_i \cup P_{a+\ell+i}$ is an s_i - t_i path. These paths are pairwise disjoint and disjoint from every M_j . Hence G is (a, ℓ) -woven. It remains to show that G_3 is $(a, 0)$ -woven.

Step 7. Three disjoint high-chromatic subgraphs: Let $R' = \{r'_1, \dots, r'_a\} \subseteq V(G_3)$. Since $\delta(G_3) \geq 21a$, we can choose pairwise distinct vertices $s'_1, \dots, s'_{2a} \in V(G_3) \setminus R'$ such that s'_i and s'_{a+i} are neighbours of r'_i . Let $S' := \{s'_1, \dots, s'_{2a}\}$ and $G_4 := G_3 - (R' \cup S')$. Then $\chi(G_4) \geq \chi(G_3) - 3a$, G_4 is an induced subgraph of G , and $\text{had}(G_4) < h$. Let $m := h(1 + \Phi)$. If G_4 were m -chromatic-inseparable, then $(\mathcal{H}_{\Phi, a_0})$ applied with $b = h \geq a_0$ would give $m \leq h\Phi$, which is false. So G_4 has vertex-disjoint induced subgraphs H_1, H_2 with $\chi(H_i) \geq \chi(G_4) - m$. The same argument applied to H_1 gives vertex-disjoint induced subgraphs $H_{1,1}, H_{1,2}$ of H_1 with $\chi(H_{1,j}) \geq \chi(H_1) - m$. Let $J_1 := H_{1,1}$, $J_2 := H_{1,2}$ and $J_3 := H_2$. Then $\chi(J_i) \geq \chi(G_4) - 2m$ for $i \in \{1, 2\}$, and $\chi(J_3) \geq \chi(G_4) - m$. By Steps 3–5, and since $2m \leq \frac{92}{3}(a+1)(1+\Phi) \leq \frac{230}{3}(1+\Phi)a$,

$$\chi(J_i) \geq \chi(G) - (92.5a + 24.5) - 42a - 3a - 2m \geq 700a \geq \lceil 3.0625(21a - 1) \rceil$$

for each i (so all these subgraphs are non-null). By Corollary 4.3, each J_i contains a $21a$ -connected induced subgraph J'_i with $\chi(J'_i) \geq \chi(J_i) - 42a$.

Step 8. Accounting: Table 3 lists the losses in chromatic number from G to J'_i , for both schemes. Collecting the losses for the uneven scheme, and using $92.5a + 24.5 + 42a + 3a + 42a \leq 179.5(a+1)$ and $m \leq \frac{46}{3}(a+1)(1+\Phi)$, we obtain

$$\chi(J'_i) \geq \chi(G) - L_i, \quad \text{where } L_i := \begin{cases} (179.5 + \frac{92}{3}(1+\Phi))(a+1) & \text{if } i \in \{1, 2\}, \\ (179.5 + \frac{46}{3}(1+\Phi))(a+1) & \text{if } i = 3. \end{cases}$$

Let $a'_1 = a'_2 := \lceil 3a/5 \rceil$ and $a'_3 := \lceil 4a/5 \rceil$. Then $a'_i < a$ (as $a \geq 9$), $a - a'_1 = \lfloor 2a/5 \rfloor \geq (2a - 4)/5$ and $a - a'_3 = \lfloor a/5 \rfloor \geq (a - 4)/5$. Since $\Lambda = \Lambda_{7.2}^U = 5(1 + \eta)(179.5 + \frac{46}{3}(1 + \Phi))$,

$$\Lambda(a - a'_3) \geq (1 + \eta)(a - 4)(179.5 + \frac{46}{3}(1 + \Phi)) \geq L_3$$

and

$$\Lambda(a - a'_1) \geq (1 + \eta)(a - 2)(359 + \frac{92}{3}(1 + \Phi)) \geq L_1 = L_2,$$

because $(1 + \eta)(a - 4) \geq a + 1$ whenever $a \geq 4 + 5/\eta$. Therefore $\chi(J'_i) \geq D(\frac{46}{3}a_0 + 4) + 1 + \Lambda a'_i$ for each i . Since J'_i is $21a$ -connected, it is $21a'_i$ -connected, and by induction each J'_i is $(a'_i, \lceil 10a'_i/3 \rceil)$ -woven.

Step 9. Weaving: Let $y := \lfloor 2a/5 \rfloor$, and partition $[a] = I_{12} \cup I_{13} \cup I_{23}$ with $|I_{13}| = |I_{23}| = y$ and $|I_{12}| = a - 2y$. Then

$$|I_{12}| + |I_{13}| = |I_{12}| + |I_{23}| = a - y = a'_1 = a'_2 \quad \text{and} \quad |I_{13}| + |I_{23}| = 2y \leq a'_3.$$

For each $i \in I_{jk}$ with $j < k$, choose terminals $t_i^{(1)} \in V(J'_j)$ and $t_i^{(2)} \in V(J'_k)$, all distinct. For $j \in \{1, 2, 3\}$, let $T_j \subseteq V(J'_j)$ consist of the terminals in J'_j together with further vertices of J'_j , so that $|T_j| = a'_j$.

Since $G_3 - R'$ is $20a$ -connected, it is $2a$ -linked by Theorem 4.5. So it contains a linkage $\mathcal{P}^{(0)}$ consisting of a path from s'_i to $t_i^{(1)}$ and a path from s'_{a+i} to $t_i^{(2)}$, for each $i \in [a]$. We now apply Lemma 4.8 three times, in the graph $G_3 - R'$. For $j = 1, 2, 3$ in turn, apply it with $H = J'_j$, $R = T_j$ and linkage $\mathcal{P}^{(j-1)}$. This is possible because J'_j is $(a'_j, 2a)$ -woven and $v(J'_j) > 21a \geq a'_j + 4a$. Here $(a'_j, 2a)$ -wovenness follows from $(a'_j, \lceil 10a'_j/3 \rceil)$ -wovenness by Lemma 4.1, since $\lceil 10a'_j/3 \rceil \geq \frac{10}{3} \cdot \frac{3a}{5} = 2a$ and $v(J'_j) > 21a \geq \frac{23}{3}a + 2 \geq a'_j + 2\lceil 10a'_j/3 \rceil$. We obtain a linkage $\mathcal{P}^{(j)}$ for the same pairs and a $K_{a'_j}$ model $\mathcal{M}_j$ in J'_j rooted at T_j , with

$$V(\mathcal{P}^{(j)}) \cap V(\mathcal{M}_j) \subseteq T_j \quad \text{and} \quad V(\mathcal{P}^{(j)}) \subseteq V(J'_j) \cup V(\mathcal{P}^{(j-1)}).$$

Let $\mathcal{P}'' := \mathcal{P}^{(3)}$. For $j < j'$ we have $V(\mathcal{P}^{(j')}) \subseteq V(\mathcal{P}^{(j)}) \cup V(J'_{j+1}) \cup \dots \cup V(J'_{j'})$, and the graphs $J'_{j+1}, \dots, J'_{j'}$ are disjoint from $\mathcal{M}_j$. Hence $V(\mathcal{P}'') \cap V(\mathcal{M}_j) \subseteq T_j$ for each j . Also all these linkages lie in $G_3 - R'$.

For $i \in I_{jk}$, let B_i be the union of $\{r'_i\}$, the vertex sets of the two paths of $\mathcal{P}''$ starting at s'_i and s'_{a+i} , the branch set of $\mathcal{M}_j$ containing $t_i^{(1)}$, and the branch set of $\mathcal{M}_k$ containing $t_i^{(2)}$. Each $G[B_i]$ is connected, because r'_i is adjacent to s'_i and s'_{a+i} . The sets B_i are pairwise disjoint, for four reasons:

- The paths of $\mathcal{P}''$ are pairwise disjoint.
- The models $\mathcal{M}_j$ lie in the disjoint graphs J'_j .
- The paths meet the models only in vertices of the sets T_j . Each branch set of $\mathcal{M}_j$ contains exactly one vertex of T_j . So a path meeting the branch set containing a terminal $t_{i'}^{(1)}$ or $t_{i'}^{(2)}$ meets it only in that terminal, and is therefore the path ending there.
- $r'_i \notin V(\mathcal{P}'')$.

Any two of the pairs $\{1, 2\}, \{1, 3\}, \{2, 3\}$ intersect. So any two of the sets B_i contain distinct branch sets of a common model $\mathcal{M}_j$, and are therefore adjacent. Hence $\{B_i\}$ is a K_a model rooted at R' , and G_3 is $(a, 0)$ -woven. This completes the proof for the uneven scheme.

The even scheme: Suppose now that $\Lambda = \Lambda_{7.2}^E$. The proof is the same, with $\ell := 3a$ and $h := 14a$, and with the following changes in the numbers. In Step 1, $|Z| = 7a$, at most $4a$ new vertices are chosen, and fewer than $11a$ vertices are forbidden. In Step 2, if $a \leq a_0$ then $\chi(G) \leq D(14a) \leq D(\frac{46}{3}a_0 + 4)$, and so $a > a_0 \geq 4 + 5/\eta \geq 9$. In Step 3, H_0 is $4a$ -linked, $v(H_0) > 40a$ and $\chi(H_0) \leq 50a + 1$. In Step 4, $2|Z| = 14a$, and the colouring bound is $28a$. In Step 5, $\chi(G_2) \geq \chi(G) - (85a + 1)$. In Step 7, $m = 14a(1 + \Phi)$, and each of J_1, J_2, J_3 satisfies

$$\chi(J_i) \geq \chi(G_4) - 2m \geq \chi(G) - (130a + 1) - 28(1 + \Phi)a \geq 300a.$$

In Step 8, $\chi(J'_i) \geq \chi(G) - (172a + 1) - 28(1 + \Phi)a \geq \chi(G) - (172.5 + 28(1 + \Phi))a$ for each i , since $a \geq 2$, and we take $a'_1 = a'_2 = a'_3 := \lceil 2a/3 \rceil$. Then $a - a'_i \geq (a - 2)/3$, and

$$\Lambda_{7.2}^E(a - a'_i) \geq (1 + \eta)(a - 2)(172.5 + 28(1 + \Phi)) \geq (172.5 + 28(1 + \Phi))a,$$

because $(1 + \eta)(a - 2) \geq a$. By induction, each J'_i is $(a'_i, 3a'_i)$ -woven. In Step 9, write $a = 3q + r_0$ with $r_0 \in \{0, 1, 2\}$, and let I_{12}, I_{13}, I_{23} have sizes (q, q, q) , $(q + 1, q, q)$ or $(q + 1, q + 1, q)$ according as $r_0 = 0, 1, 2$. Then each of the three sums $|I_{12}| + |I_{13}|$, $|I_{12}| + |I_{23}|$ and $|I_{13}| + |I_{23}|$ is at most $\lceil 2a/3 \rceil$. Each J'_j is $(a'_j, 2a)$ -woven by Lemma 4.1, since $2a \leq 3a'_j$ and $v(J'_j) > 21a \geq 7a'_j$, and $v(J'_j) \geq a'_j + 4a$. The rest of the proof is unchanged. $\square$

The proportions of the uneven scheme minimise the coefficient of Φ for this method. Suppose the three subgraphs host fractions $\alpha_1, \alpha_2, \alpha_3$ of the branch sets, so that $\alpha_1 + \alpha_2 + \alpha_3 = 2$ since each branch set uses two of them. Suppose also that the lemma proves $(a, \beta a)$ -wovenness. Then we need $\beta \alpha_i \geq 2$ for each i , the Hadwiger bound h at which Step 7 separates is about $(2 + 4\beta)a$, and the coefficient of Φ is about $(2 + 4\beta) \max\{2/(1 - \alpha_1), 2/(1 - \alpha_2), 1/(1 - \alpha_3)\}$, since J'_3 costs one separation and J'_1, J'_2 cost two. A short optimisation over the weights of the three pairs gives the minimum $\frac{46}{3} \cdot 5 = \frac{230}{3}$, at $\alpha_1 = \alpha_2 = \frac{3}{5}$, $\alpha_3 = \frac{4}{5}$ and $\beta = \frac{10}{3}$. Delcourt and Postle use $\alpha_i = \frac{2}{3}$ and $\beta = 3$, which gives $14 \cdot 6 = 84$.

Theorem 7.3. *Let $t \geq 2$ be an integer, and let $\eta \in (0, 1]$, $a_0 \geq 4 + 5/\eta$ and $\Phi \geq 0$ be real. If G satisfies $\text{had}(G) < t$ and $(\mathcal{H}_{\Phi, a_0})$, then*

$$\chi(G) < D(\frac{46}{3}a_0 + 4) + 1 + 42t + (1 + \eta) \min\{517.5 + 84(1 + \Phi), 897.5 + \frac{230}{3}(1 + \Phi)\} t.$$

Operation	Uneven scheme	Even scheme
Delete Z and the paths $\mathcal{P}_1$ (Steps 3–5)	$5 Z \leq \frac{115}{3}a + 10$	$35a$
Delete the hub H_0 (Step 3)	$12.5(a + \ell) + 2 \leq 54.17a + 14.5$	$50a + 1$
Subtotal (Step 5)	$92.5a + 24.5$	$85a + 1$
Extract G_3 (Step 5)	$42a$	$42a$
Delete $R' \cup S'$ (Step 7)	$3a$	$3a$
Chromatic separations (Step 7)	$2m, 2m, m$ for J_1, J_2, J_3	$2m$ for each J_i
Extract J'_i (Step 7)	$42a$	$42a$
Total L_i (Step 8)	$179.5(a + 1) + 2m$ ($i = 1, 2$) $179.5(a + 1) + m$ ($i = 3$)	$172a + 1 + 2m$
where $m = h(1 + \Phi)$ and	$h \leq \frac{46}{3}(a + 1)$	$h = 14a$
Sizes a'_1, a'_2, a'_3	$\lceil 3a/5 \rceil, \lceil 3a/5 \rceil, \lceil 4a/5 \rceil$	$\lceil 2a/3 \rceil$ each

Table 3: The losses in chromatic number in the proof of Lemma 7.2, where $\ell = \lceil 10a/3 \rceil$ in the uneven scheme and $\ell = 3a$ in the even scheme, and $|Z| = a + 2\ell$.

Theorem 7.3 parallels a theorem of Norin and Steiner [36, Theorem 15 and Appendix B.3], and the derivation by Delcourt and Postle of their Theorem 1.6 from their Theorem 7.1 [8].

Proof. Suppose not, and let Λ be as in Lemma 7.2. Then $\chi(G) \geq \lceil 3.0625(21t - 1) \rceil$. By Corollary 4.3, G has a $21t$ -connected induced subgraph G' with $\chi(G') \geq \chi(G) - 42t \geq D(\frac{46}{3}a_0 + 4) + 1 + \Lambda t$. The graph G' satisfies $(\mathcal{H}_{\Phi, a_0})$. By Lemma 7.2 with $a = t$, G' has a K_t minor. This contradicts $\text{had}(G) < t$. $\square$

Corollary 7.4. *For every $\Phi \geq 0$ and $\xi > 0$ there is $t_0 = t_0(\Phi, \xi) \geq 40$ such that the following holds for every integer $t \geq t_0$. Let G be a graph with $\text{had}(G) < t$ such that every s -chromatic-inseparable induced subgraph H of G with $\text{had}(H) < b$, for some real $b \geq t/\log t$, satisfies $s \leq \Phi b$. Then $\chi(G) \leq (c_{7.4}(\Phi) + \xi)t$, where $c_{7.4}(x) := 42 + \min\{517.5 + 84(1 + x), 897.5 + \frac{230}{3}(1 + x)\}$.*

Proof. Since $t \geq 40$ and $x/\log x$ is increasing for $x \geq e$, we have $a_0 := t/\log t \geq 40/\log 40 > 9$. Let $\eta := 5/(a_0 - 4) \in (0, 1]$, so that $a_0 = 4 + 5/\eta$ and G satisfies $(\mathcal{H}_{\Phi, a_0})$. By Theorem 4.11, $D(\frac{46}{3}a_0 + 4) + 1 = O(t/\sqrt{\log t})$. Moreover $\eta(897.5 + \frac{230}{3}(1 + \Phi)) \rightarrow 0$ as $t \rightarrow \infty$, and the bound of Theorem 7.3 is at most $D(\frac{46}{3}a_0 + 4) + 1 + (c_{7.4}(\Phi) + \eta(897.5 + \frac{230}{3}(1 + \Phi)))t$. The claim follows from Theorem 7.3. $\square$

8 An additive bootstrap

Let $\beta > 0$ and $c \geq 0$. We say that c is *admissible for exponent β* if for every $\xi > 0$ there is t_ξ such that, for every real $t \geq t_\xi$, every graph G with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^\beta$ satisfies $\chi(G) \leq (c + \xi)t$.

By Theorem 4.15, 4 is admissible for every exponent $\beta < \frac{1}{2}$.

Theorem 8.1. *Let $\beta, \beta_0 > 0$ and $c, c_0 \geq 0$ be real, and let $0 < \zeta < \beta_0/2$. Suppose that c_0 is admissible for every exponent in $(0, \beta_0)$, and that c is admissible for exponent β . Then $2c + c_{7.4}(c_0)$ is admissible for exponent $\beta + \zeta$, where $c_{7.4}$ is as in Corollary 7.4.*

Proof. The proof follows the proof of a theorem of Norin and Steiner [36, Theorem 5] in their Section 3.2. The main change is that the Gyárfás graphs are coloured using c_0 rather than c . We also allow every $b \geq t/\log t$, use Corollary 7.4 in place of their Theorem 15, pass to real t by rounding t up, and treat edgeless H separately. Note that $c_{7.4}(c_0 + \xi) \leq c_{7.4}(c_0) + 84\xi$, since $c_{7.4}$ is the minimum of two affine functions with slopes

84 and $\frac{230}{3}$. Fix $\xi > 0$ and β' with $2\zeta < \beta' < \beta_0$, and let t be sufficiently large. Let G be a graph with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^{\beta+\zeta}$, and let $k := (\log t)^\zeta$. Choose a maximal family $H_1, \dots, H_m$ of pairwise vertex-disjoint connected induced bipartite subgraphs of G , each with more than k vertices. Let $G_1 := G[V(H_1) \cup \dots \cup V(H_m)]$ and $G_2 := G - V(G_1)$. By maximality, $\text{bip}(G_2) \leq k$.

Colouring G_1 . The sets $V(H_i)$ partition $V(G_1)$ into sets inducing connected subgraphs of chromatic number at most 2. Let G'_1 be obtained from G_1 by contracting each H_i to a vertex. Then G'_1 is a minor of G , so $\text{had}(G'_1) < t$. Also $v(G'_1) = m \leq v(G)/k \leq t(\log t)^\beta$. By Lemma 4.16 and the admissibility of c , $\chi(G_1) \leq 2\chi(G'_1) \leq 2(c + \xi)t$.

Colouring G_2 . Let H be an induced subgraph of G_2 that is s -chromatic-inseparable and satisfies $\text{had}(H) < b$ for some real $b \geq t/\log t$, and let $b' := \min\{b, t\}$. Then $\text{had}(H) < b'$, as $\text{had}(H) \leq \text{had}(G) < t$, and $\log b' \geq \log t - \log \log t$. We claim that $s \leq (c_0 + \xi)b$.

If $E(H) = \emptyset$, then $s < \chi(H) \leq 1 \leq (c_0 + \xi)b$, since $b \geq t/\log t \geq 1/\xi$ for t large. Otherwise $\text{bip}(H) \leq \text{bip}(G_2) \leq k$, since H is an induced subgraph of G_2 . Since $\text{had}(H) < b'$, Lemma 4.17 (with $a = b'$) shows that some graph J with $\text{had}(J) < b'$ and $v(J) \leq 2b'k^2$ satisfies $\chi(J) > s$. For t sufficiently large, $2b'k^2 = 2b'(\log t)^{2\zeta} \leq b'(\log b')^{\beta'}$, because $2\zeta < \beta'$ and $\log b' \geq \frac{1}{2} \log t$. Since $b' \geq t/\log t$ is also large, the admissibility of c_0 for exponent β' , applied with b' in place of t , gives $\chi(J) \leq (c_0 + \xi)b'$. Hence $s < (c_0 + \xi)b' \leq (c_0 + \xi)b$, as claimed.

Since $x \mapsto x/\log x$ is increasing for $x \geq e$, every $b \geq \lceil t \rceil / \log \lceil t \rceil$ satisfies $b \geq t/\log t$. So Corollary 7.4 applies with $\Phi := c_0 + \xi$ and with $\lceil t \rceil$ in place of t , and gives $\chi(G_2) \leq (c_{7.4}(c_0) + 85\xi)\lceil t \rceil \leq (c_{7.4}(c_0) + 86\xi)t$ for t large. Therefore $\chi(G) \leq \chi(G_1) + \chi(G_2) \leq (2c + c_{7.4}(c_0) + 88\xi)t$. Since $\xi > 0$ was arbitrary, $2c + c_{7.4}(c_0)$ is admissible for exponent $\beta + \zeta$. $\square$

Corollary 8.2. *For every integer $L \geq 0$, the number*

$$c^{(L)} := 2^L \cdot 4 + (2^L - 1) \cdot c_{7.4}(4) = 2^L \cdot 4 + (2^L - 1) \cdot 979.5$$

is admissible for every exponent $\beta \in (0, \frac{1}{2} + \frac{L}{4})$. In particular, $c^{(7)} = 124,908.5$ is admissible for every exponent in $(0, 2.25)$.

Proof. We use induction on L . The case $L = 0$ is Theorem 4.15. Suppose the claim holds for L , and let $\beta < \frac{1}{2} + \frac{L+1}{4}$. Choose ζ with $\max\{0, \beta - \frac{1}{2} - \frac{L}{4}\} < \zeta < \min\{\frac{1}{4}, \beta\}$. This interval is non-empty, since $\beta < \frac{1}{2} + \frac{L+1}{4}$ and $\beta > 0$. Then $0 < \zeta < \frac{1}{4}$ and $0 < \beta - \zeta < \frac{1}{2} + \frac{L}{4}$. By induction, $c^{(L)}$ is admissible for exponent $\beta - \zeta$. Theorem 8.1 (with $c_0 = 4$ and $\beta_0 = \frac{1}{2}$) shows that $2c^{(L)} + c_{7.4}(4) = c^{(L+1)}$ is admissible for exponent β . $\square$

9 Chromatic-inseparable graphs

The following proposition is a version of a lemma of Delcourt and Postle [8, Lemma 2.5] in which the small-graph term enters additively.

Proposition 9.1. *Let $\omega_{6.2} = 0.0044$ and $\tau_{6.2} = 4.56 \cdot 10^{18}$ be the constants of Theorem 6.2, and let $C_{5.4} = 70$ and $M = 10^5$ be the constants of Lemma 5.4. Let $C_{9.1} := 18,000$, so that $\omega_{6.2}C_{9.1} \geq C_{5.4} + 9$. Let t be an integer with $\log t \geq 1.6 \cdot 10^6$ (so in particular $t \geq 10^{13}$). Let*

$$N_t := 24\tau_{6.2}M^2 t \log^2 t.$$

Let H be a graph with no K_t minor, and let $\Gamma \geq 0$ be a real number such that every subgraph of H with at most N_t vertices has chromatic number at most Γ . Let

$$\sigma := 78t + 2\Gamma \quad \text{and} \quad \Psi := (32 + 2C_{9.1})t + 2\Gamma + 2.$$

If $\chi(H) \geq \Psi$, then H is σ -chromatic-separable. Consequently, if H is s -chromatic-inseparable for some real $s \geq 0$, then $s < (2C_{9.1} + 33)t + 2\Gamma$.

The two conclusions differ in an important way. The separation cost σ does not involve the large constant $C_{9.1}$, which enters only through the chromatic threshold Ψ , the lower bound on $\chi(H)$ in the first conclusion. The lemma of Delcourt and Postle [8, Lemma 2.5] has the shape of the first conclusion with $\Psi = 2\sigma$, so their large constant appears in both the separation cost and the chromatic threshold. The second conclusion is the single-parameter form, which matches hypothesis $(\mathcal{H}_{\Phi, a_0})$. In the final application (Section 10.4) we use the first conclusion.

Proof. The proof follows the proof of a lemma of Delcourt and Postle [8, Lemma 6.4]. The statement $(*)_T$ below is the inductive statement of that proof, Step 1 is the argument of their Corollary 6.1 [8], and the iterated use of Lemma 4.8 in Step 4 is their Claim 6.4.1 [8]. Steps 2, 3, 6 and 7 follow the corresponding parts of their proof: the Menger paths to the hub, the extraction of H_3 , the re-optimisation by their Lemma 6.2, and the final case distinction. Our indices p_j, p'_j and $q_{i,j}$ play the roles of their indices b_j, b'_j and $a_{i,j}$. We use Corollary 4.3 in place of their Theorem 5.1, Theorem 6.2 in place of their Theorem 2.3, and Lemmas 4.6 and 5.4 in place of their Theorem 5.8 and Lemma 5.13 [8]. We use M times as many stages as they do, each adding M times fewer branch sets, so that the rooted cliques in the pieces are small. We record the chromatic number lost at each step, so that the small-graph term is never multiplied by an absolute constant. Step 5 is new: we treat the first stage separately, since for $T = 1$ the construction of Delcourt and Postle gives no adjacency between the new branch sets.

Parameters. Let $x := \lceil t/(M\sqrt{\log t}) \rceil$ and $r := \lceil M\sqrt{\log t} \rceil$, so that $rx \geq t$. Let $k := 12t$ and $d := C_{9.1}t$. Here k is the connectivity of the high-chromatic subgraphs extracted by Corollary 4.3, and d is the density threshold: Theorem 6.2 is applied to subgraphs with at least d times as many edges as vertices. Define

$$\Theta := 6t + \Gamma + 2k, \quad \sigma = 30t + 2\Gamma + 4k, \quad \Psi = 8t + 2\Gamma + 2k + 2d + 2.$$

The number Γ need not be an integer, and the proof uses only that it is an upper bound on the chromatic number of the subgraphs of H with at most N_t vertices.

Reduction. It suffices to prove the following claim:

(†) if H is σ -chromatic-inseparable and $\chi(H) \geq \Psi$, then H has a K_t minor.

Indeed, since H has no K_t minor, (†) shows that if $\chi(H) \geq \Psi$ then H is σ -chromatic-separable. For the second conclusion, suppose H is s -chromatic-inseparable with $s \geq \max\{\sigma, \Psi\}$. Then H is σ -chromatic-inseparable, and $\chi(H) > s \geq \Psi$, a contradiction. Hence $s < \max\{\sigma, \Psi\} \leq (2C_{9.1} + 33)t + 2\Gamma$, since $\sigma = 78t + 2\Gamma$ and $\Psi = (32 + 2C_{9.1})t + 2\Gamma + 2$.

Bounds on orders. For $1 \leq T \leq r$ we have $T \leq M\sqrt{\log t} + 1$, and $(M\sqrt{\log t} + 1)^2 \leq 4M^2 \log t$. Hence the numbers $r\tau_{6.2}t \log t$, $T\tau_{6.2}t \log t$ and $6T^2\tau_{6.2}t \log t$ are all at most N_t . By the choice of Γ , every subgraph of H with at most that many vertices has chromatic number at most Γ .

Applicability of Corollary 4.3. Each application of Corollary 4.3 below extracts a k -connected induced subgraph from a graph with chromatic number at least $\Psi - \Theta - 24t - \Gamma = 2d - 22t + 2$. This is at least $\lceil 3.0625(k - 1) \rceil$, because $d \geq 1.8 \cdot 10^4 t$ and $k = 12t$. Each application loses at most $2k$ in chromatic number.

Induction. Assume that H is σ -chromatic-inseparable, $\chi(H) \geq \Psi$, and H has no K_t minor. We prove by induction on $T \in \{0, 1, \dots, r\}$ the following statement:

$(*)_T$: H has a K_{Tx} model $\mathcal{A} = \{A_1, \dots, A_{Tx}\}$ and a set $S \subseteq V(H)$ with the following five properties.

- (I1) S is a core of $\mathcal{A}$, and $|S| \leq 2T^2\tau_{6.2}t \log t$.
- (I2) H has a k -connected subgraph J^* .
- (I3) J^* is tangent to $\mathcal{A}$, that is, $|V(J^*) \cap A_j| = 1$ for every j .
- (I4) $V(J^*) \cap V(\mathcal{A}) \subseteq S$.
- (I5) $\chi(J^*) \geq \chi(H) - \Theta$.

Since $rx \geq t$, the statement $(*)_r$ gives a K_t minor of H . This contradicts our assumption, and so proves $(\dagger)$. The subgraph J^* need not be induced. In the inductive step it is the source of high chromatic number, and its tangency to $\mathcal{A}$ lets us attach the new branch sets to the old ones.

The case $T = 0$. Since $\chi(H) \geq \Psi \geq \lceil 3.0625(k-1) \rceil$, Corollary 4.3 shows that H has a k -connected induced subgraph J^* with $\chi(J^*) \geq \chi(H) - 2k \geq \chi(H) - \Theta$. Take $\mathcal{A}$ to be the empty model and $S := \emptyset$. Then (I1)–(I5) hold, (I1), (I3) and (I4) vacuously.

The inductive step. Let $1 \leq T \leq r$ and assume $(*)_{T-1}$. Let $\mu' := (T-1)x$ and $\mu := Tx$. Then $\mu' \leq t + M\sqrt{\log t} \leq 1.0001t$ and $\mu \leq 3t$. Moreover $r \leq M\sqrt{\log t} + 1$ and $x \leq t/(M\sqrt{\log t}) + 1$, so

$$\begin{aligned} 3\mu' + \mu &= (4T-3)x \leq (4M\sqrt{\log t} + 1) \left(\frac{t}{M\sqrt{\log t}} + 1 \right) \\ &= 4t + \frac{t}{M\sqrt{\log t}} + 4M\sqrt{\log t} + 1 \leq 4.004t, \end{aligned}$$

since $\log t \geq 1.6 \cdot 10^6$. As $3\mu' + \mu$ is an integer, $3\mu' + \mu \leq \lfloor 4.004t \rfloor$. Let $\mathcal{A}' = \{A'_1, \dots, A'_{\mu'}\}$, S' and $H' := J^*$ be given by $(*)_{T-1}$. By (I3), for $j \in [\mu']$ there is a unique vertex w_j in $V(H') \cap A'_j$. Since the sets A'_j are disjoint, the vertices $w_1, \dots, w_{\mu'}$ are distinct, and

$$V(H') \cap V(\mathcal{A}') = \{w_1, \dots, w_{\mu'}\} \subseteq S'. \quad (1)$$

The inclusion holds by (I4). Every subgraph of H' that avoids $\{w_1, \dots, w_{\mu'}\}$ is therefore disjoint from $V(\mathcal{A}')$.

Step 1. Small pieces: Let $H_1 := H' - \{w_1, \dots, w_{\mu'}\}$. Then $\chi(H_1) \geq \chi(H') - \mu' \geq \chi(H) - \Theta - 2t$. Choose a maximal family of pairwise vertex-disjoint $(9t+1)$ -connected subgraphs J of H_1 such that $v(J) \leq \tau_{6.2}t \log t$ and $e(J) > (\omega_{6.2}d - 9t)v(J)$.

Suppose for a contradiction that this family has fewer than r members, and let U be the union of their vertex sets. Then $|U| \leq r\tau_{6.2}t \log t \leq N_t$, so $\chi(H[U]) \leq \Gamma$. Let F be any non-null subgraph of $H_1 - U$. If $e(F) \geq dv(F)$, then Theorem 6.2 with $9t$ in place of k gives a subgraph of F that could be added to the family, contradicting maximality. Theorem 6.2 applies because F has no K_t minor, $t \geq 10^{13}$, $d = C_{9.1}t \geq 1.8 \cdot 10^4 t$ and $9t \leq \omega_{6.2}d/2$. So $e(F) < dv(F)$, and F has a vertex of degree less than $2d$. Hence $H_1 - U$ is $(2d-1)$ -degenerate and $\chi(H_1 - U) \leq 2d$. Then $\chi(H_1) \leq 2d + \Gamma$. But $\chi(H_1) \geq \Psi - \Theta - 2t = 2d + \Gamma + 2$, a contradiction.

So the family has at least $r \geq T$ members. Call T of them $J_0, J_1, \dots, J_{T-1}$. We call these graphs the *pieces*, and J_0 the *hub*. These are pairwise vertex-disjoint subgraphs of H_1 , each with at most $\tau_{6.2}t \log t$ vertices. Each piece J is $9t$ -connected and satisfies $e(J) > (\omega_{6.2}d - 9t)v(J) \geq C_{5.4}t v(J)$. So $v(J) > 2e(J)/v(J) > 2C_{5.4}t > k$, and by Lemma 5.4, J is $(2x, \lfloor 4.004t \rfloor)$ -woven. By (1), they are disjoint from $V(\mathcal{A}')$.

Step 2. Paths to the hub J_0 : If $T = 1$, let $W_1 := \emptyset$, $\mathcal{Q}_1 := \emptyset$ and $\bar{J} := V(J_0)$. Now let $T \geq 2$. For $i \in [T-1]$, choose $2x$ vertices of J_i and label them $w_{\mu'+2(i-1)x+1}, \dots, w_{\mu'+2ix}$. Let X_i be this set, and let $W_1 := \{w_1, \dots, w_{3\mu'}\}$. Then $|W_1| = 3\mu' \leq 6t$. The vertices $w_1, \dots, w_{\mu'}$ lie in H' but not in H_1 , while each X_i lies in $J_i \subseteq H_1$. So the $3\mu'$ labelled vertices are distinct, and $W_1 \subseteq V(H')$ is disjoint from $V(J_0)$. We now argue as in Step 4 of the proof of Lemma 7.2. Let H'^+ be obtained from H' by adding, for each $w \in W_1$,

a new vertex w^+ adjacent to w and to every neighbour of w in H' . Each new vertex has more than k neighbours, so H'^+ is k -connected, and hence $2|W_1|$ -connected since $2|W_1| = 6\mu' \leq 12t = k$. The sets $W_1 \cup \{w^+ : w \in W_1\}$ and $V(J_0)$ are disjoint, the first has $2|W_1|$ vertices, and the second has more than $k \geq 2|W_1|$ vertices. By Theorem 4.4 with $2|W_1|$ in place of k , H'^+ has $2|W_1|$ pairwise vertex-disjoint paths from $W_1 \cup \{w^+ : w \in W_1\}$ to $V(J_0)$, each meeting these two sets only in its ends. A path starting at w^+ does not contain w , so its second vertex is a neighbour of w in H' . Replacing w^+ by w gives a W_1 - $V(J_0)$ path in H' . Finally, replace each path by a shortest path between its ends in the subgraph of H' induced by its vertex set. We obtain a family $\mathcal{Q}_1$ of $2|W_1|$ W_1 - $V(J_0)$ paths in H' , each an induced path of H' . These paths are pairwise disjoint outside W_1 , and each vertex of W_1 is an end of exactly two of them. Each path minus its first vertex is an induced path of H' , so it can be coloured with two colours of its own. Hence the subgraph of H' induced by $V(\mathcal{Q}_1) \setminus W_1$ has chromatic number at most $2 \cdot 2|W_1| = 12\mu'$. Let $\bar{J} := V(J_0) \cup V(J_1) \cup \dots \cup V(J_{T-1})$. Then $|\bar{J}| \leq T\tau_{6,2} t \log t \leq N_t$, so $\chi(H[\bar{J}]) \leq \Gamma$.

Step 3. The next high-chromatic subgraph: Let $H_2 := H' - (\bar{J} \cup V(\mathcal{Q}_1) \cup W_1)$. The vertices removed are those of $\bar{J}$, of $V(\mathcal{Q}_1) \setminus W_1$, and of $W_1 \setminus \bar{J} = \{w_1, \dots, w_{\mu'}\}$. Their removal costs at most Γ , $12\mu'$ and μ' colours respectively. Since $13\mu' \leq 13(t + M\sqrt{\log t}) \leq 24t$, we obtain $\chi(H_2) \geq \chi(H') - 24t - \Gamma$. For $T = 1$ the loss is only Γ . By Corollary 4.3, H_2 has a k -connected induced subgraph H_3 with

$$\chi(H_3) \geq \chi(H) - \Theta - 24t - \Gamma - 2k = \chi(H) - \sigma.$$

Here Corollary 4.3 applies because $\chi(H_2) \geq \Psi - \Theta - 24t - \Gamma$. By construction, H_3 is disjoint from $\bar{J}$, from $V(\mathcal{Q}_1)$ and from W_1 . By (1), it is also disjoint from $V(\mathcal{A}')$.

Step 4. Building a K_μ model when $T \geq 2$:

A linkage to the hub. The graph $H' - W_1$ is 2μ -connected, as $k - |W_1| \geq 12t - 6t \geq 2\mu$. The sets $V(H_3)$ and $V(J_0)$ are disjoint, avoid W_1 , and have more than $k \geq 2\mu$ vertices each. So, by Theorem 4.4, $H' - W_1$ has a family $\mathcal{Q}_2$ of 2μ pairwise disjoint $V(H_3)$ - $V(J_0)$ paths. Let their first vertices be $a_1, b_1, \dots, a_\mu, b_\mu$. Let H'' be obtained from $H'[V(\mathcal{Q}_1) \cup V(\mathcal{Q}_2) \cup V(J_0)]$ by identifying a_i and b_i to a new vertex c_i , for each i . Let $C := \{c_1, \dots, c_\mu\}$. In H'' the hypotheses of Lemma 4.7 hold with $A_1 = W_1$, $A_2 = C$, and $V(J_0)$ in place of B :

- $\mathcal{Q}_1$ avoids C , because it avoids H_3 . So $\mathcal{Q}_1$ survives the identification as a family of W_1 - $V(J_0)$ paths in $H'' - C$.
- $\mathcal{Q}_2$ avoids W_1 .
- The paths of $\mathcal{Q}_2$ are pairwise disjoint outside C , and each c_i lies in exactly two of them.

So Lemma 4.7 gives $3\mu' + \mu$ pairwise disjoint $(W_1 \cup C)$ - $V(J_0)$ paths in H'' , one starting at each vertex of $W_1 \cup C$. We undo the identification as follows. A path of this family starting at c_i contains no other vertex of $W_1 \cup C$. Its second vertex is adjacent in H'' to c_i , and hence adjacent in H' to a_i or to b_i . Replace c_i by such a vertex, and call it $w_{3\mu'+i}$. All other edges of the path are edges of H' . This yields pairwise vertex-disjoint paths $P_1, \dots, P_{3\mu'+\mu}$ in H' , where P_i joins w_i to a vertex $u_i \in V(J_0)$, and $w_{3\mu'+i} \in \{a_i, b_i\}$ for $i \in [\mu]$. The vertices $w_1, \dots, w_{3\mu'+\mu}$ and $u_1, \dots, u_{3\mu'+\mu}$ are $2(3\mu' + \mu)$ distinct vertices. So $\mathcal{P} = \{P_1, \dots, P_{3\mu'+\mu}\}$ is a $\{(w_i, u_i)\}$ -linkage in H' . Every path of $\mathcal{Q}_1 \cup \mathcal{Q}_2$ meets $V(H_3) \cup V(J_0)$ only in its ends. Moreover, three facts hold. First, the only vertices of H_3 among those used to build H'' are the vertices a_i, b_i , since $\mathcal{Q}_1$ and J_0 avoid H_3 and each path of $\mathcal{Q}_2$ meets H_3 only in its first vertex. Second, each path given by Lemma 4.7 meets $W_1 \cup C$ only in its first vertex. Third, $V(H') \cap V(\mathcal{A}') = \{w_1, \dots, w_{\mu'}\} \subseteq W_1$. Hence

$$\begin{aligned} V(\mathcal{P}) \cap V(H_3) &= \{w_{3\mu'+1}, \dots, w_{3\mu'+\mu}\}, & V(\mathcal{P}) \cap V(J_0) &= \{u_1, \dots, u_{3\mu'+\mu}\}, \\ V(\mathcal{P}) \cap V(\mathcal{A}') &= \{w_1, \dots, w_{\mu'}\}. \end{aligned}$$

Rooted models in the pieces. Each J_i is $(2x, \lfloor 4.004t \rfloor)$ -woven, by Step 1. Since $v(J_i) > C_{5.4}t \geq 2x + 2\lfloor 4.004t \rfloor$ and $3\mu' + \mu \leq \lfloor 4.004t \rfloor$, Lemma 4.1 shows that J_i is $(2x, 3\mu' + \mu)$ -woven. The same inequality gives the hypothesis $v(J_i) \geq 2x + 2(3\mu' + \mu)$ of Lemma 4.8. Let $\mathcal{P}^{(0)} := \mathcal{P}$. For $i = 1, \dots, T-1$ in turn, apply Lemma 4.8 in the graph H' , with the linkage $\mathcal{P}^{(i-1)}$, the subgraph J_i of $H_1 \subseteq H'$ and the root set X_i . This gives a $\{(w_l, u_l)\}$ -linkage $\mathcal{P}^{(i)}$ in H' and a K_{2x} model $\mathcal{M}_i$ in J_i rooted at X_i , with

$$V(\mathcal{P}^{(i)}) \cap V(\mathcal{M}_i) \subseteq X_i \quad \text{and} \quad V(\mathcal{P}^{(i)}) \subseteq V(J_i) \cup V(\mathcal{P}^{(i-1)}).$$

Let $\mathcal{P}' := \mathcal{P}^{(T-1)}$. For $i < i'$, the second inclusion gives $V(\mathcal{P}^{(i')}) \subseteq V(\mathcal{P}^{(i)}) \cup V(J_{i+1}) \cup \dots \cup V(J_{i'})$, and the graphs $J_{i+1}, \dots, J_{i'}$ are disjoint from $\mathcal{M}_i \subseteq J_i$. Hence $V(\mathcal{P}') \cap V(\mathcal{M}_i) \subseteq X_i$. Since every vertex of X_i is the first vertex of a path of every linkage $\mathcal{P}^{(l)}$, in fact $V(\mathcal{P}') \cap V(\mathcal{M}_i) = X_i$. Also $V(\mathcal{P}') \subseteq V(\mathcal{P}) \cup V(J_1) \cup \dots \cup V(J_{T-1})$. The graphs J_i are disjoint from H_3 , from $V(\mathcal{A}')$ and from J_0 . So the three displayed intersection properties of $\mathcal{P}$ also hold for $\mathcal{P}'$. For $w \in X_i$, let $B(w)$ denote the branch set of $\mathcal{M}_i$ containing w , and let P'_i denote the path of $\mathcal{P}'$ starting at w_i .

Indices. For $i \in [T-1]$ and $j \in [x]$, let $q_{i,j} := \mu' + 2(i-1)x + x + j$, so that the vertices $w_{q_{i,j}}$ form the second half of X_i . For $j \in [\mu']$, let $i(j) := \lceil j/x \rceil$, let $p_j := \mu' + 2(i(j)-1)x + (j - (i(j)-1)x)$, and let $p'_j := 3\mu' + j$. The vertices w_{p_j} form the first halves of the sets X_i . Define

$$S_j := \{u_j, u_{p_j}, u_{p'_j}\} \quad (j \in [\mu']), \quad S_{\mu'+j} := \{u_{q_{i,j}} : i \in [T-1]\} \cup \{u_{4\mu'+j}\} \quad (j \in [x]).$$

The index sets $\{j : j \in [\mu']\}$, $\{p_j : j \in [\mu']\}$, $\{p'_j : j \in [\mu']\}$, $\{q_{i,j}\}$ and $\{4\mu' + j : j \in [x]\}$ partition $[3\mu' + \mu]$. So $S_1, \dots, S_\mu$ are pairwise disjoint and contain $3\mu' + \mu$ vertices in total. Table 4 lists, for each new branch set, the paths of $\mathcal{P}'$ that it uses to reach the hub J_0 .

Branch set	Path	First vertex	Lies in
$A''_j, j \in [\mu']$	P'_j	w_j	W_1 , the old branch set A'_j
	P'_{p_j}	w_{p_j}	first half of $X_{i(j)}$, in $J_{i(j)}$
	$P'_{p'_j}, p'_j = 3\mu' + j$	$w_{3\mu'+j}$	H_3
$A''_{\mu'+j}, j \in [x]$	$P'_{q_{i,j}}, i \in [T-1]$	$w_{q_{i,j}}$	second half of X_i , in J_i
	$P'_{4\mu'+j}$	$w_{4\mu'+j}$	H_3

Table 4: The paths of $\mathcal{P}'$ used by each branch set of $\mathcal{A}''$ in Step 4 of the proof of Proposition 9.1. The last vertex of each path lies in S_j or $S_{\mu'+j}$, respectively, and hence in D_j or $D_{\mu'+j}$.

Knitting in the hub. The graph J_0 is $9t$ -connected, and $e(J_0) > C_{5.4}t v(J_0) \geq 5(3\mu' + \mu) v(J_0)$. Moreover $2(3\mu' + \mu) - \mu = 7\mu' + x \leq 7(t + M\sqrt{\log t}) + t/(M\sqrt{\log t}) + 1 \leq 7.001t \leq 9t$. By Lemma 4.6 with $3\mu' + \mu$ and μ in place of k and ℓ , the graph J_0 is $(3\mu' + \mu, \mu)$ -knot, so there are pairwise disjoint sets $D_1, \dots, D_\mu \subseteq V(J_0)$ with $S_j \subseteq D_j$ and each $J_0[D_j]$ connected.

The model. For $j \in [\mu']$ let

$$A''_j := A'_j \cup V(P'_j) \cup D_j \cup V(P'_{p_j}) \cup B(w_{p_j}) \cup V(P'_{p'_j}),$$

and for $j \in [x]$ let

$$A''_{\mu'+j} := V(P'_{4\mu'+j}) \cup D_{\mu'+j} \cup \bigcup_{i \in [T-1]} (V(P'_{q_{i,j}}) \cup B(w_{q_{i,j}})).$$

Each path in these definitions joins a vertex of A'_j , of some set $B(\cdot)$, or of H_3 , to its end in the corresponding set D_j . So each $H[A''_j]$ is connected. The sets A''_j are pairwise disjoint, for five reasons:

- The sets A'_j , the paths of $\mathcal{P}'$, the sets D_j and the sets $B(\cdot)$ are each pairwise disjoint.
- $\mathcal{P}'$ meets $V(\mathcal{A}')$, $V(J_0)$ and the models $\mathcal{M}_i$ only in path ends. Each such end lies in the same set A''_j as its path.
- J_0 and the graphs J_i are disjoint from $V(\mathcal{A}')$.
- J_0 is disjoint from the graphs J_i .
- Every element of S_j belongs to D_j .

The sets are pairwise adjacent. Two sets $A''_j, A''_{j'}$ with $j, j' \leq \mu'$ are adjacent because $\mathcal{A}'$ is a model. For $j \leq \mu'$ and $j' \in [x]$, the sets A''_j and $A''_{\mu'+j'}$ contain the distinct branch sets $B(w_{p_j})$ and $B(w_{q_{i(j),j'}})$ of $\mathcal{M}_{i(j)}$. For distinct $j, j' \in [x]$, the sets $A''_{\mu'+j}$ and $A''_{\mu'+j'}$ contain the distinct branch sets $B(w_{q_{1,j}})$ and $B(w_{q_{1,j'}})$ of $\mathcal{M}_1$. So $\mathcal{A}'' := \{A''_1, \dots, A''_\mu\}$ is a K_μ model.

Tangency and core. The sets A'_j , D_j and $B(\cdot)$ are disjoint from H_3 , by Step 3 and since J_0 and the graphs J_i lie in $\bar{J}$. By the intersection properties of $\mathcal{P}'$, the paths of $\mathcal{P}'$ meet $V(H_3)$ exactly in the vertices $w_{3\mu'+1}, \dots, w_{3\mu'+\mu}$, and $w_{3\mu'+i}$ is the first vertex of $P'_{3\mu'+i}$. The index $3\mu' + i$ is used by exactly one set A''_j , namely $j = i$, since $p'_i = 3\mu' + i$ for $i \leq \mu'$ and the index $4\mu' + (i - \mu')$ belongs to A''_i for $i > \mu'$. So H_3 meets A''_j exactly in $w_{3\mu'+j}$, which lies in S as defined below. Hence H_3 is tangent to $\mathcal{A}''$. Let $S := S' \cup \bar{J} \cup \{w_{3\mu'+1}, \dots, w_{3\mu'+\mu}\}$. Adjacencies between the sets A''_j with $j \leq \mu'$ are witnessed by edges with both ends in S' , and all other adjacencies by edges inside some $J_i \subseteq \bar{J}$. Hence S is a core of $\mathcal{A}''$, and $V(H_3) \cap V(\mathcal{A}'') \subseteq S$. Finally,

$$|S| \leq 2(T-1)^2 \tau_{6.2} t \log t + T \tau_{6.2} t \log t + 3t \leq 2T^2 \tau_{6.2} t \log t,$$

because $2(T-1)^2 + T + 1 \leq 2T^2$ for $T \geq 1$ and $3t \leq \tau_{6.2} t \log t$.

Step 5. Building a K_μ model when $T = 1$: Here $\mu' = 0$, there are no pieces J_i with $i \geq 1$, and $\mu = x$. The construction in Step 4 does not apply, because it uses the model $\mathcal{M}_1$ to make the new branch sets pairwise adjacent. Instead, since H' is x -connected and $V(H_3)$ and $V(J_0)$ are disjoint sets with more than $k \geq x$ vertices each, Theorem 4.4 gives x pairwise disjoint $V(H_3)$ – $V(J_0)$ paths $P_1^\circ, \dots, P_x^\circ$ in H' , with P_j° joining $y_j \in V(H_3)$ to $z_j \in V(J_0)$. The graph J_0 is $(2x, \lfloor 4.004t \rfloor)$ -woven, by Step 1. Since $v(J_0) > C_{5.4}t \geq 2x + 2\lfloor 4.004t \rfloor$, Lemma 4.1 shows that J_0 is $(x, 0)$ -woven. So J_0 has a K_x model $\{B_1, \dots, B_x\}$ with $z_j \in B_j \subseteq V(J_0)$. Let $A''_j := V(P_j^\circ) \cup B_j$ and $S := V(J_0) \cup \{y_1, \dots, y_x\}$. Since each P_j° meets $V(J_0)$ only in z_j and $V(H_3)$ only in y_j , these sets form a K_x model to which H_3 is tangent. Also S is a core, since all required edges lie in J_0 , and $|S| \leq \tau_{6.2} t \log t + x \leq 2\tau_{6.2} t \log t$.

Step 6. Re-optimisation: For each $j \in [\mu]$, apply Lemma 4.10 to the connected graph $H[A''_j]$ and the set $S \cap A''_j$, which is non-empty since it contains the vertex of H_3 in A''_j . We obtain a set $A_j \subseteq A''_j$ with $H[A_j]$ connected, and a set X'_j with $S \cap A''_j \subseteq X'_j \subseteq A_j$, $|X'_j| \leq 3|S \cap A''_j|$ and $\chi(H[A_j \setminus X'_j]) \leq 2$. We check that $\mathcal{A} := \{A_1, \dots, A_\mu\}$ is a K_μ model with core S . The sets $A_j \subseteq A''_j$ are pairwise disjoint, and each $H[A_j]$ is connected. Since $S \cap A''_j \subseteq A_j \subseteq A''_j$, we have $A_j \cap S = A''_j \cap S$. As S is a core of $\mathcal{A}''$, for all distinct j, j' some edge joins $A''_j \cap S = A_j \cap S$ to $A''_{j'} \cap S = A_{j'} \cap S$. So $\mathcal{A}$ is a K_μ model with core S . Next, H_3 meets each A''_j in exactly one vertex, and that vertex lies in S , as shown in Steps 4 and 5. So it lies in A_j . Hence $|V(H_3) \cap A_j| = 1$ for every j , that is, H_3 is tangent to $\mathcal{A}$. Also $V(H_3) \cap V(\mathcal{A}) \subseteq V(H_3) \cap V(\mathcal{A}'') \subseteq S$. Let $A := \bigcup_j A_j$ and $X' := \bigcup_j X'_j$. Then $|X'| \leq 3|S| \leq 6T^2 \tau_{6.2} t \log t \leq N_t$, so $\chi(H[X']) \leq \Gamma$. Also $A \setminus X'$ is the disjoint union of the sets $A_j \setminus X'_j$, and each $H[A_j \setminus X'_j]$ can be coloured with two colours of its own. So $\chi(H[A \setminus X']) \leq 2\mu \leq 6t$. Hence $\chi(H[A]) \leq \chi(H[X']) + \chi(H[A \setminus X']) \leq 6t + \Gamma$.

Step 7. Restoring the invariant: Let $H_4 := H - A$. Then H_4 is an induced subgraph of H , and

$$\chi(H_4) \geq \chi(H) - \chi(H[A]) \geq \chi(H) - 6t - \Gamma \geq \Psi - 6t - \Gamma \geq \lceil 3.0625(k-1) \rceil.$$

By Corollary 4.3, H_4 has a k -connected induced subgraph H_5 with $\chi(H_5) \geq \chi(H) - 6t - \Gamma - 2k = \chi(H) - \Theta$. Since H_4 is an induced subgraph of H , so is H_5 . Also $V(H_5) \cap A = \emptyset$.

Case 1. $|V(H_5) \cap V(H_3)| < k$: Let $Y_1 := V(H_5) \setminus V(H_3)$ and $Y_2 := V(H_3)$. These sets are disjoint. So $H[Y_1]$ and $H[Y_2]$ are vertex-disjoint induced subgraphs of H . Deleting a vertex lowers the chromatic number by at most 1, and $H[Y_1] = H_5 - (V(H_5) \cap V(H_3))$ since H_5 is induced. Hence

$$\chi(H[Y_1]) \geq \chi(H_5) - (k - 1) \geq \chi(H) - \Theta - k \geq \chi(H) - \sigma,$$

since $\sigma - \Theta = 24t + \Gamma + 2k \geq k$. Moreover H_3 is a spanning subgraph of $H[Y_2]$, so $\chi(H[Y_2]) \geq \chi(H_3) \geq \chi(H) - \sigma$ by Step 3. Thus H is σ -chromatic-separable, a contradiction. So Case 1 does not occur.

Case 2. $|V(H_5) \cap V(H_3)| \geq k$: Let $J^* := H_3 \cup H_5$, a subgraph of H . We claim that J^* is k -connected. Indeed, $v(J^*) \geq v(H_5) > k$. Let $Z \subseteq V(J^*)$ with $|Z| < k$. Then $H_3 - Z$ and $H_5 - Z$ are connected, because H_3 and H_5 are k -connected. They share a vertex, because $|V(H_3) \cap V(H_5)| \geq k > |Z|$. So $J^* - Z = (H_3 - Z) \cup (H_5 - Z)$ is connected. This proves the claim.

We now verify $(*)_T$ for $\mathcal{A}$, S and J^* . Property (I1) holds by Step 6, and by the bound on $|S|$ in Step 4 or Step 5. Property (I2) was just shown. For (I3) and (I4), recall that $V(H_5) \cap A = \emptyset$. So $V(J^*) \cap A_j = V(H_3) \cap A_j$ for every j . This set has exactly one vertex, by Step 6. Also $V(J^*) \cap V(\mathcal{A}) = V(H_3) \cap V(\mathcal{A}) \subseteq S$, by Step 6. Finally, $\chi(J^*) \geq \chi(H_5) \geq \chi(H) - \Theta$, which is (I5). This proves $(*)_T$. $\square$

The small-graph quantity Γ entered the proof only in Steps 1, 2–3 and 6, each time with coefficient 1. It appears once in Θ and twice in each of σ and Ψ , while the large constant $C_{9.1}$ enters only through the term $2d$ of Ψ (Step 1), and $k = 12t$ contributes only $O(t)$.

10 Proof of Theorem 1.3

In this section we make every size threshold explicit and prove Theorem 1.3. Here a *size threshold* is a lower bound on t (or on a) above which a statement holds, such as $\log \log t \geq 881$. Below the size threshold of the bootstrap we colour using Lemma 10.6, and below the size threshold of the final step (Case 1 of the proof of Theorem 1.3) we use Theorem 4.11. We use the base constant 4.1 in place of $4 + \varepsilon$, and we allow an additive slack of 1 at each of the seven bootstrapping steps. All non-trivial numerical inequalities in this paper were checked by computer, and the scripts are available as ancillary files with the arXiv version of this paper. There are three kinds of check. First, the final constants, and the other combinations of rational constants such as $c_{10.4}$, $c_{10.5}$ and $c_{1.3}$, were checked in exact rational arithmetic (`precise_check.py`). Second, inequalities involving logarithms, square roots and exponentials at specific values were checked in interval arithmetic with outward rounding at 60 digits (`precise_check.py`, using the `mpmath` library), so these checks are certified despite rounding errors. Third, since numbers such as $\exp(e^{881})$ are far too large to handle directly, each inequality required for all t above a size threshold is rewritten in terms of $x := \log \log t$, so that $\log t = e^x$, and reduced, by monotonicity in x , to its value at the size threshold, which is then checked as above. The monotonicity is immediate in most cases, and is proved in the text where it is not. The scripts `explicit_thresholds.py` and `refined_lin_check.py` use floating-point arithmetic, and only locate the size thresholds and summarise the computations; they are not needed for the proof. The dynamic program mentioned in Appendix A is `bootstrap_dp.py`.

10.1 An explicit small-graph theorem

We use the following two lemmas of Norin and Steiner.

Lemma 10.1 (Norin and Steiner [36, Lemma 7]). *Let $n \in \mathbb{N}$, $m \geq 2$ and $\gamma \in (0, 1)$ be such that $n \geq (600(m+2)/\gamma)^{20m+40}$. For every graph G on n vertices with $\alpha(G) \leq m$, there is a graph Z on $V(G)$ with $\chi(G) \leq \chi_f(G \cup Z) + \gamma n$ and $\Delta(Z) \leq (600(m+2)/\gamma)^{20m+41}$.*

Lemma 10.2 (Norin and Steiner [36, Lemma 8]). *Let $m, d, t \geq 1$ be real and let G be a graph. If $\text{had}(G) < t$, $\alpha(G) \leq m$, and Z is a graph on $V(G)$ with $\Delta(Z) \leq d$, then $\chi_f(G \cup Z) \leq 4t + (d+1)^{4m}$.*

Lemma 10.3. *Let a be real with $\log \log a \geq 879.8$. Every graph J with $\text{had}(J) < a$ and $v(J) \leq a(\log a)^{0.49}$ satisfies $\chi(J) \leq 4.1a$.*

Proof. We make explicit the proof by Norin and Steiner of their Theorem 4 from their Lemmas 7 and 8 [36]. Let $\Upsilon := (\log a)^{0.49}$, $m := \lceil 25\Upsilon \rceil - 1$ and $\gamma := 0.04/\Upsilon$. Then $2 \leq m \leq 25\Upsilon$ and $\gamma \in (0, 1)$. While the current graph has an independent set of size at least $m+1$, remove one and give it a new colour. This uses at most $v(J)/(m+1) \leq 0.04a$ colours. It leaves an induced subgraph J' with $\alpha(J') \leq m$ and $n' := v(J') \leq a\Upsilon$. Let $q := 600(m+2)/\gamma$, so that $q \leq 600 \cdot 25\Upsilon(25\Upsilon+2)$. Let $d := q^{20m+41}$. We now show that $\log \log a \geq 879.8$ implies

$$(20m+40) \log q \leq \log a \quad \text{and} \quad 4m \log(d+1) \leq \log(0.02a).$$

Let $x := \log \log a \geq 879.8$, so that $\log a = e^x$ and $\Upsilon = e^{0.49x} \geq e^{431}$. Let $y := \log 375,000 + 0.98x$. Since $q \leq 375,000\Upsilon^2(1 + 0.08/\Upsilon)$ and $\log(1+u) \leq u$, we have $\log q \leq y + 0.08/\Upsilon$. Since $m \leq 25\Upsilon$ and $\Upsilon \geq e^{431}$,

$$(20m+40) \log q \leq (500\Upsilon+40)(y+0.08/\Upsilon) \leq 501\Upsilon y = \exp(0.49x + \log(501y)).$$

The function $0.51x - \log(501y)$ is increasing for $x \geq 879.8$, since its derivative is $0.51 - 0.98/y > 0$. At $x = 879.8$ it equals $448.7 - 13.0 > 0$. So $(20m+40) \log q \leq e^x = \log a$, which is the first inequality. For the second, $\log(d+1) \leq (20m+41) \log q + \log 2$, and so

$$4m \log(d+1) \leq 100\Upsilon(500\Upsilon+41)(y+0.08/\Upsilon) + 100\Upsilon \log 2 \leq 50,000 \Upsilon^2 y (1 + 0.1/\Upsilon).$$

On the other hand, $\log(0.02a) = e^x - \log 50 \geq e^x(1 - 4e^{-x})$. Taking logarithms, and using $\log(1+u) \leq u$ and $-\log(1-u) \leq 2u$ for $0 \leq u \leq \frac{1}{2}$, the second inequality follows from

$$F(x) := \log 50,000 + 0.98x + \log(\log 375,000 + 0.98x) \leq x - 0.1e^{-431} - 8e^{-x}.$$

The right side exceeds $x - 10^{-180}$. At $x = 879.8$ we have $F(x) < 879.799$, which is less than $x - 10^{-180}$. Moreover $F(x) - x$ is decreasing, since its derivative is $0.98 + 0.98/y - 1 < 0$. So the second inequality holds for every $x \geq 879.8$.

If $n' < q^{20m+40}$, then $\chi(J') \leq n' < q^{20m+40} \leq a$, by the first inequality. Otherwise Lemma 10.1 applies. Together with Lemma 10.2 (with a in place of t), it gives

$$\chi(J') \leq 4a + (d+1)^{4m} + \gamma n' \leq 4a + 0.02a + 0.04a.$$

In either case $\chi(J) \leq 0.04a + 4.06a = 4.1a$. □

10.2 Explicit bootstrapping

In Lemma 10.4 below, the exponents β_j increase by ζ at each step, starting from $\beta_0 = 0.49 < \frac{1}{2}$. The constant $c_{10.4}$ there equals $c_{7.4}(4.1) + 1 = 42 + 517.5 + 84 \cdot 5.1 + 1 = 988.9$, with $c_{7.4}$ as in Corollary 7.4.

Lemma 10.4. *Let $\zeta := 0.2425$, let $\beta_j := 0.49 + j\zeta$ for $j \geq 0$, and let $c_{10.4} := 988.9$. Let $j \geq 1$ be an integer and $c \geq 0$ be real. Suppose that, for every real t with $\log \log t \geq 881$, every graph G with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^{\beta_{j-1}}$ satisfies $\chi(G) \leq ct$. Then, for every real t with $\log \log t \geq 881$, every graph G with $\text{had}(G) < t$ and $v(G) \leq t(\log t)^{\beta_j}$ satisfies $\chi(G) \leq (2c + c_{10.4})t$.*

Proof. Follow the proof of Theorem 8.1 with $k := (\log t)^\zeta$. Use the hypothesis of the lemma in place of the admissibility of c . Use Lemma 10.3 in place of the admissibility of c_0 . We check the three places where t must be large.

- (i) *Colouring G_1 .* The minor G'_1 has at most $v(G)/k \leq t(\log t)^{\beta_{j-1}}$ vertices. So $\chi(G_1) \leq 2ct$.
- (ii) *The Gyárfás graphs.* Let H be an s -chromatic-inseparable induced subgraph of G_2 with $\text{had}(H) < b$ for some real $b \geq t/\log t$, and let $b' := \min\{b, t\}$, as in the proof of Theorem 8.1. If $E(H) = \emptyset$, then $s < \chi(H) \leq 1 \leq 4.1b$. So assume that $E(H) \neq \emptyset$. Then $\log b' \geq \log t - \log \log t \geq e^{879.8}$, so $\log \log b' \geq 879.8$. The inequality $2(\log t)^{2\zeta} \leq (\log t - \log \log t)^{\beta_0}$ holds, since it requires only $\log \log t \geq 138.7$. So every Gyárfás graph J in this application of Lemma 4.17, that is, every graph J with $\text{had}(J) < b'$ and $v(J) \leq 2b'k^2$, satisfies $v(J) \leq 2b'k^2 \leq b'(\log b')^{0.49}$, and so $\chi(J) \leq 4.1b'$ by Lemma 10.3 with b' in place of a . On the other hand, $\text{had}(H) < b'$, $E(H) \neq \emptyset$ and $\text{bip}(H) \leq k$. So, by Lemma 4.17 with $a = b'$, the hypothesis of that lemma fails for s , that is, some Gyárfás graph J satisfies $\chi(J) > s$. Hence $s < 4.1b' \leq 4.1b$, and so G_2 satisfies $(\mathcal{H}_{4.1, a_0})$ with $a_0 := t/\log t$.
- (iii) *The chromatic-separable reduction.* Apply Theorem 7.3 to G_2 , with $t' := \lceil t \rceil$ in place of t , $a_0 = t/\log t$, $\eta := 5/(a_0 - 4)$ and $\Phi = 4.1$. Let $s_0 := \lfloor \frac{46}{3}a_0 + 4 \rfloor \leq 16a_0$. By Theorem 4.11, $D(\frac{46}{3}a_0 + 4) + 1 \leq 60s_0\sqrt{\log s_0} + 2 \leq 960a_0\sqrt{\log(16t)} + 2 \leq t/4$, since $\log \log t \geq 16.6$. Since $t' \leq t + 1$, and $\eta \leq 10^{-6}$ because $a_0 - 4 \geq 5 \cdot 10^6$,

$$\chi(G_2) \leq \frac{t}{4} + (42 + 945.9)(t + 1) + 945.9\eta(t + 1) \leq 988.9t = c_{10.4}t.$$

Hence $\chi(G) \leq \chi(G_1) + \chi(G_2) \leq (2c + c_{10.4})t$. $\square$

Proposition 10.5. *Let $c_{10.5} := 2^7 \cdot 4.1 + (2^7 - 1) \cdot c_{10.4} = 126,115.1$, where $c_{10.4} = 988.9$ is as in Lemma 10.4. Let $N_a := 24\tau_{6.2}M^2 a \log^2 a$, where $\tau_{6.2} = 4.56 \cdot 10^{18}$ and $M = 10^5$ are as in Theorem 6.2 and Lemma 5.4, so that N_a is the quantity N_t of Proposition 9.1 with a in place of t . For every real a with $\log \log a \geq 881$, every graph J with $\text{had}(J) < a$ and $v(J) \leq N_a$ satisfies $\chi(J) \leq c_{10.5}a$.*

Proof. By Lemma 10.3, the hypothesis of Lemma 10.4 holds for $j = 1$ with $c = 4.1$. Apply Lemma 10.4 seven times. The constants satisfy $\bar{c}_0 = 4.1$ and $\bar{c}_j = 2\bar{c}_{j-1} + c_{10.4}$, so $\bar{c}_7 = c_{10.5}$. Hence every graph J with $\text{had}(J) < a$ and $v(J) \leq a(\log a)^{\beta_7}$ satisfies $\chi(J) \leq c_{10.5}a$, where $\beta_7 = 2.1875$. Finally, $N_a \leq a(\log a)^{2.1875}$ whenever $(\log a)^{0.1875} \geq 24\tau_{6.2}M^2$. This holds whenever $\log \log a \geq \log(24\tau_{6.2}M^2)/0.1875$, and $\log(24\tau_{6.2}M^2)/0.1875 < 369$. $\square$

10.3 Small scales

Lemma 10.6. *Let $N_a = 24\tau_{6.2}M^2 a \log^2 a$ be as in Proposition 10.5. For every integer $a \geq 3$, every graph J with no K_a minor and $v(J) \leq N_a$ satisfies*

$$\chi(J) \leq (6 + 2\log(24\tau_{6.2}M^2) + 4\log \log a)a \leq (144.4 + 4\log \log a)a.$$

Proof. This is the proof by Delcourt and Postle of their Theorem 1.5 in their Section 3 [8], with N_a in place of their order bound. Let $p := 2(a - 1)$. By Theorem 4.13, every non-null subgraph J' of J has no K_a minor, and so satisfies $\alpha(J') \geq v(J')/p$. Since $N_a \geq p$, Lemma 4.14 gives

$$\chi(J) \leq (2 + \log(N_a/p))p \leq 2a \left(2 + \log(12\tau_{6.2}M^2 \log^2 a) + \log \frac{a}{a-1} \right).$$

Since $\log \frac{a}{a-1} \leq \log \frac{3}{2} < 1$ and $\log 12 < \log 24$, the right side is at most $(6 + 2 \log(24\tau_{6.2}M^2) + 4 \log \log a)a$. Finally, $2 \log(24\tau_{6.2}M^2) < 138.4$. $\square$

Let $a \geq 3$ be an integer with $\log \log a < 881$. Then, by Lemma 10.6, every graph J with no K_a minor and $v(J) \leq N_a$ satisfies $\chi(J) \leq (144.4 + 4 \cdot 881)a < 3700a \leq c_{10.5}a$. Together with Proposition 10.5, we obtain the following.

Corollary 10.7. *Let $c_{10.5} = 126,115.1$ and N_a be as in Proposition 10.5. For every integer $a \geq 3$, every graph J with no K_a minor and $v(J) \leq N_a$ satisfies $\chi(J) \leq c_{10.5}a$.*

10.4 The final separation step

Proposition 9.1 and Corollary 10.7 combine as follows.

Corollary 10.8. *Let b be an integer with $\log b \geq 1.6 \cdot 10^6$, and let*

$$\sigma_b := (78 + 2c_{10.5})b \quad \text{and} \quad \Psi_b := (32 + 2C_{9.1} + 2c_{10.5})b + 2.$$

Every graph F with no K_b minor and $\chi(F) \geq \Psi_b$ is σ_b -chromatic-separable.

Proof. Every subgraph of F with at most N_b vertices has no K_b minor, and so has chromatic number at most $c_{10.5}b$, by Corollary 10.7. So Proposition 9.1 applies with b , F and $c_{10.5}b$ in place of t , H and Γ , and its first conclusion is the claim. $\square$

In the final application, we use Corollary 10.8 in Step 7 of the proof of Lemma 7.2, in place of $(\mathcal{H}_{\Phi, a_0})$. The point is that the separation cost σ_b does not involve $C_{9.1}$. The chromatic threshold Ψ_b does, but it is at most a fixed multiple of t , and it can be paid once, as an additive term that is carried through the induction.

Lemma 10.9. *Let t be an integer with $\log \log t \geq 25$, and let $a_0 := t/\log t$ and $\eta := 5/(a_0 - 4)$. Let*

$$\Lambda_{10.9} := (1 + \eta) \left(897.5 + \frac{230}{3}(78 + 2c_{10.5}) \right) \quad \text{and} \quad \Lambda'_{10.9} := 110 + 2C_{9.1} + 4c_{10.5} + \frac{2}{t}.$$

Let $a \geq 1$ be an integer, and let G be a $21a$ -connected graph with no K_t minor and

$$\chi(G) \geq D\left(\frac{46}{3}a_0 + 4\right) + 1 + \Lambda_{10.9}a + \Lambda'_{10.9}t.$$

Then G is $(a, \lceil 10a/3 \rceil)$ -woven. In particular, G has a K_a minor.

Proof. Since $a_0 \geq 9$, we have $\eta \in (0, 1]$ and $a_0 = 4 + 5/\eta$. The proof is the proof of Lemma 7.2 for the uneven scheme, with the following changes. Throughout, t is fixed, and, as in the proof of Lemma 7.2, we use induction on a . Every graph to which the induction hypothesis is applied is a subgraph of G , so it has no K_t minor, and the term $\Lambda'_{10.9}t$ is carried unchanged through the induction, as the term Ct is in the proofs of Delcourt and Postle [8, Theorem 7.1] and Norin and Steiner [36, Lemma B.12]. Steps 1–6 and 9 are unchanged, and Step 2 gives $a > a_0$ as before, since the hypothesis on $\chi(G)$ is now stronger. Let $x^* := 78 + 2c_{10.5}$. In Step 7, the choice of R' and S' , the definition of G_4

and the bound $\chi(G_4) \geq \chi(G_3) - 3a$ are unchanged. The rest of Step 7 up to the definition of J_1, J_2, J_3 is replaced by the following, and Step 8 is modified as described below.

The graph G_4 is a subgraph of G , and it has no K_h minor by Step 2. Let $h' := \min\{h, t\}$, which is an integer. Then G_4 has no $K_{h'}$ minor. Also $h' \geq \min\{a, t\} > a_0$, so $\log h' \geq \log t - \log \log t \geq e^{25} - 25 \geq 1.6 \cdot 10^6$. Since $h' \leq t$,

$$\Psi_{h'} + \sigma_{h'} = (110 + 2C_{9.1} + 4c_{10.5})h' + 2 \leq \Lambda'_{10.9}t.$$

By Steps 3–5, and since $\Lambda_{10.9}a \geq 137.5a + 24.5$, we have $\chi(G_4) \geq \chi(G) - (137.5a + 24.5) \geq \Lambda'_{10.9}t \geq \Psi_{h'} + \sigma_{h'}$. By Corollary 10.8 with $b = h'$, the graph G_4 is $\sigma_{h'}$ -chromatic-separable. So it has vertex-disjoint induced subgraphs H_1, H_2 with $\chi(H_i) \geq \chi(G_4) - \sigma_{h'} \geq \Psi_{h'}$. Applying Corollary 10.8 to H_1 , which also has no $K_{h'}$ minor, gives vertex-disjoint induced subgraphs $H_{1,1}, H_{1,2}$ of H_1 with $\chi(H_{1,j}) \geq \chi(H_1) - \sigma_{h'}$. Let $J_1 := H_{1,1}$, $J_2 := H_{1,2}$ and $J_3 := H_2$, and let $m := \sigma_{h'}$. Then $m \leq x^*h \leq \frac{46}{3}x^*(a+1)$ and $2m \leq \frac{230}{3}x^*a$. So the rest of Step 7, and Step 8, go through as in the proof of Lemma 7.2 with $1 + \Phi$ replaced by x^* , with $\Lambda_{10.9}$ in place of $\Lambda_{7.2}^U$, and with the additional term $\Lambda'_{10.9}t$ on both sides. In particular, each J'_i is a $21a'_i$ -connected graph with no K_t minor and

$$\chi(J'_i) \geq D\left(\frac{46}{3}a_0 + 4\right) + 1 + \Lambda_{10.9}a'_i + \Lambda'_{10.9}t,$$

so the induction hypothesis applies to it. $\square$

The even scheme would give $3(1 + \eta)(172.5 + 28x^*)$ in place of $\Lambda_{10.9}$, which is larger, since $84 > \frac{230}{3}$ and x^* is large.

10.5 Completing the proof

Proof of Theorem 1.3. Let

$$c'_{1.3} := 7030.6 + 2C_{9.1} + \frac{472}{3}c_{10.5} \quad \text{and} \quad c_{1.3} := (1 + 10^{-6})c'_{1.3}.$$

Numerically, $c'_{1.3} = 19,885,139\frac{2}{3}$ and $c_{1.3} < 19,885,160$.

Let G be a graph. We may assume that G has an edge, since otherwise $\chi(G) \leq \text{had}(G)$, as both are 0 if G is null and both are 1 if G is non-null and edgeless. Let $h := \text{had}(G) \geq 2$ and $t := h + 1$, so that G has no K_t minor. We first show that $\chi(G) \leq c'_{1.3}t$ or $\chi(G) \leq 1.7 \cdot 10^7t$, according to the size of t . Then we convert this into a bound in terms of h .

Case 1. $\log \log t < 25$: By Theorem 4.11, $\chi(G) \leq 60t\sqrt{\log t} + 1 \leq 60e^{12.5}t + 1 < 1.7 \cdot 10^7t$.

Case 2. $\log \log t \geq 25$: Let $a_0, \eta, \Lambda_{10.9}$ and $\Lambda'_{10.9}$ be as in Lemma 10.9. We claim that

$$\chi(G) < D\left(\frac{46}{3}a_0 + 4\right) + 1 + 42t + \Lambda_{10.9}t + \Lambda'_{10.9}t.$$

Suppose not. Then $\chi(G) \geq \lceil 3.0625(21t - 1) \rceil$, so by Corollary 4.3, G has a $21t$ -connected induced subgraph G' with $\chi(G') \geq \chi(G) - 42t \geq D\left(\frac{46}{3}a_0 + 4\right) + 1 + \Lambda_{10.9}t + \Lambda'_{10.9}t$. The graph G' has no K_t minor. By Lemma 10.9 with $a = t$, it has a K_t minor, a contradiction. This proves the claim. (This is the argument of Theorem 7.3.)

Since $\log \log t \geq 25$, we have $a_0 \geq \exp(e^{25} - 25)$, which is far larger than 10^{10} . Hence $\eta(897.5 + \frac{230}{3}(78 + 2c_{10.5})) \leq \frac{1}{2}$, and also $\Lambda'_{10.9}t \leq (110 + 2C_{9.1} + 4c_{10.5} + \frac{1}{2})t$. Moreover, as in Lemma 10.4, $D\left(\frac{46}{3}a_0 + 4\right) + 1 \leq 960a_0\sqrt{\log(16t)} + 2 \leq t/10$, since $\log \log t \geq 18.4$. Therefore

$$\chi(G) \leq \left(0.1 + 42 + 897.5 + \frac{230}{3}(78 + 2c_{10.5}) + \frac{1}{2} + 110 + 2C_{9.1} + 4c_{10.5} + \frac{1}{2}\right)t = c'_{1.3}t,$$

since $0.1 + 42 + 897.5 + \frac{230}{3} \cdot 78 + 1 + 110 = 7030.6$ and $\frac{460}{3} + 4 = \frac{472}{3}$.

Conversion to a bound in terms of h . Suppose first that $h < 10^6$. Then Theorem 4.11 gives

$$\chi(G) \leq 60t\sqrt{\log t} + 1 \leq 60(h+1)\sqrt{\log(10^6)} + 1 \leq 120 \cdot 3.72h + 1 < 500h.$$

Now suppose that $h \geq 10^6$. Then $t = h + 1 \leq (1 + 10^{-6})h$. In Case 1, $\chi(G) < 1.7 \cdot 10^7(1 + 10^{-6})h < c_{1.3}h$. In Case 2, $\chi(G) \leq c'_{1.3}(1 + 10^{-6})h = c_{1.3}h$. In all cases $\chi(G) \leq c_{1.3} \text{had}(G) \leq 19,885,160 \text{had}(G)$. $\square$

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A Further improvements

We now identify the parts of the argument that determine the main constant, and indicate what would be needed to reduce it further.

The dominant term. The constant $c'_{1.3} = 7030.6 + 2C_{9.1} + \frac{472}{3}c_{10.5}$ in the proof of Theorem 1.3 is dominated by $\frac{472}{3}c_{10.5} \approx 1.984 \cdot 10^7$, about 99.8% of $c'_{1.3}$. Of this, $\frac{460}{3}c_{10.5}$ comes from the separation cost $\sigma_b \approx 2c_{10.5}b$ in Lemma 10.9, multiplied by the coefficient $\frac{230}{3}$ of the chromatic-separable reduction, and $4c_{10.5}$ comes from the chromatic threshold, through the bound $\Psi_b + \sigma_b \leq \Lambda'_{10.9}t$. The term $2C_{9.1} = 36,000$ contributes less than 0.2%. So the constant is determined almost entirely by the bootstrap constant $c_{10.5}$ and by the coefficient $\frac{230}{3}$. In particular, further improvements to the weaving constant $C_{5.4}$ or to the proportion $\omega_{6.2}$ would change the constant by less than 0.2%.

The chromatic-inseparable case. The parameters of Theorem 6.2 are chosen so that $\omega_{6.2}C_{9.1} \geq C_{5.4} + 9$ with $C_{9.1} = 18,000$, and they have not been optimised further. Lemma 5.2 needs density λ and connectivity about $2s + 2\ell \approx 8t$. Delcourt and Postle [8, Lemma 5.13] delete the roots before applying Wollan’s theorem, and obtain the density from connectivity. We apply Lemma 5.2 to the piece itself, so only its density matters. Using Wollan’s Theorem 5.1 [46] in place of Lemma 5.3 would give $C_{5.4} \approx 4893$ and would need $C_{9.1} \approx (4893 + 9)/\omega_{6.2} \approx 1.11 \cdot 10^6$, which would give a constant of about $2.208 \cdot 10^7$, about 11% more. Using Proposition 9.1 only through its second conclusion, that is, Lemma 7.2 with $(\mathcal{H}_{\Phi, a_0})$ in place of Lemma 10.9, gives a constant of about $1017.3 + \frac{230}{3}(2C_{9.1} + 33) + \frac{460}{3}c_{10.5} \approx 2.210 \cdot 10^7$, about 11% more.

The bootstrap. We do not claim that the schedule of Corollary 8.2 is optimal in any absolute sense. The proof only needs some admissible schedule, and Proposition 10.5 gives an explicit one. Within the method of Section 8, a general schedule is built as follows. Let $A(g)$ denote a constant that is admissible for every exponent in $(0, g)$, starting from $A(g) = 4$ for $g \leq \frac{1}{2}$. Theorem 8.1 gives $A(g_1 + g_0/2) \leq 2A(g_1) + c_{7.4}(A(g_0))$, since every $\beta < g_1 + g_0/2$ can be written as $\beta = \beta' + \zeta$ with $\beta' < g_1$ and $\zeta < g_0/2$. Here the Gyárfás graphs may be bounded using any previously established constant $A(g_0)$. We ran a dynamic program over this recursion, with g_0, g_1 multiples of $1/400$ (and also of

1/40 and 1/800), in exact rational arithmetic and with $c_{7.4}$ as in Corollary 7.4. A step may also be used for any smaller exponent, since a constant admissible for every exponent in $(0, g)$ is admissible for every exponent in $(0, g')$ when $g' \leq g$. Since exponent 2 itself must be covered, the target is a constant admissible for every exponent in $(0, g)$ for some $g > 2$. The smallest such constant it finds is exactly $c^{(7)}$ (see the script `bootstrap_dp.py` mentioned in Section 10). So, among such schedules on these grids, using the base bound for the Gyárfás graphs at every step is best. Using a non-base bound for them multiplies the chromatic-separability constants.

The small-graph theorem. As Norin and Steiner observe [36, Section 2.1], Theorem 4.15 stops at exponent $\frac{1}{2}$ because the error term in Lemma 10.2 is $(d+1)^{4m}$ with $d = \exp(O(m \log m))$, that is, $\exp(O(m^2 \log m))$. An error of $\exp(O(m \log m))$ would extend the base to exponent $1 - o(1)$. Then ζ could be close to $\frac{1}{2}$ in Theorem 8.1, and three bootstrapping steps would suffice instead of seven. Three steps would replace $c_{10.5}$ by $2^3 \cdot 4.1 + 7c_{10.4} = 6955.1$, and since $c_{10.5}$ dominates $c'_{1.3}$, this would reduce $c'_{1.3}$ from about $2.0 \cdot 10^7$ to about $1.1 \cdot 10^6$.

The chromatic-separable reduction. The coefficient $\frac{230}{3} = 5 \cdot \frac{46}{3}$ comes from two sources. The factor $\frac{46}{3}$ comes from separating at $h \approx \frac{46}{3}a$, which is forced by the use of Lemma 4.9 to obtain rooted $K_{|Z|}$ models from a $K_{2|Z|}$ minor, where $|Z| \approx \frac{23}{3}a$. The factor 5 comes from the uneven distribution of branch sets in Step 9 of Lemma 7.2, which minimises the coefficient of Φ for that method (see the remark after the lemma). Its constant term 897.5 is larger than the constant term 517.5 of the even distribution, so the bootstrapping uses the even scheme. Delcourt and Postle's even distribution gives $84 = 6 \cdot 14$. Rooted $K_a + \ell K_2$ models from a $K_{(a+2\ell)(1+o(1))}$ minor would halve the factor $\frac{46}{3}$. Lemma 7.1 already uses the density form of Theorem 4.5 for the hub. The linkage cost of $G_3 - R'$ in Step 9 could perhaps be reduced in the same way, but this would require density and connectivity in the same subgraph, which Corollary 4.3 does not provide. Such a reduction would lower $c_{10.4}$, and hence $c_{10.5}$, which now dominates the constant. Similarly, halving the factor $\frac{46}{3}$ would roughly halve the constant, since $\frac{230}{3}$ multiplies $2c_{10.5}$ in $\Lambda_{10.9}$. The base constant 4 (the factor 2 of Reed and Seymour times the factor 2 from a bipartition step in the proof of Lemma 10.2) contributes through the term $84(1+c_0)$ in $c_{7.4}(c_0)$ and through the term $2^L \cdot 4$ in $c^{(L)}$ (and $2^7 \cdot 4.1$ in $c_{10.5}$).